The Medium Access Control (MAC) Sublayer

Principles, Protocols, and Architectures

Dr Vijay Ukani

Based on Computer Networks by Tanenbaum

Disclaimer

- The contents presented in the slides is compilation of information from various sources. The source of information is mentioned on the slides.
- The slides are used for academic purpose. The distribution and use of the slides for academic use is permitted.

Learning outcomes

By the end, students should be able to…

Explain the MAC problem

Why collisions arise when multiple stations share a broadcast medium.

Compare access protocols

ALOHA, CSMA, CSMA/CD, collision-free and limited-contention approaches.

Reason about performance

Vulnerable period, propagation delay, load, throughput and backoff.

Explain Ethernet

Classic shared Ethernet, switched Ethernet, frame structure and evolution.

Explain Wi-Fi MAC

CSMA/CA, ACK, interframe spaces, backoff, RTS/CTS and hidden terminals.

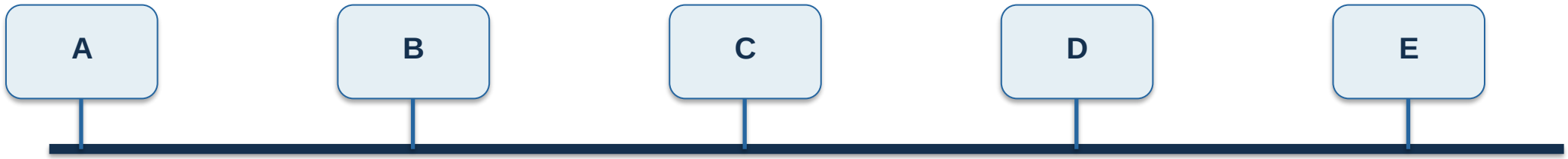
Understand Layer-2 switching

Learning bridges, forwarding, loops, spanning tree and VLANs.

- **The chapter is not a list of protocols; it is a sequence of solutions to the same shared-medium problem.**

One medium, many senders

At any instant, who is allowed to transmit?



If two transmit

Signals overlap at the receiver → collision / interference.

If nobody transmits

Capacity is wasted.

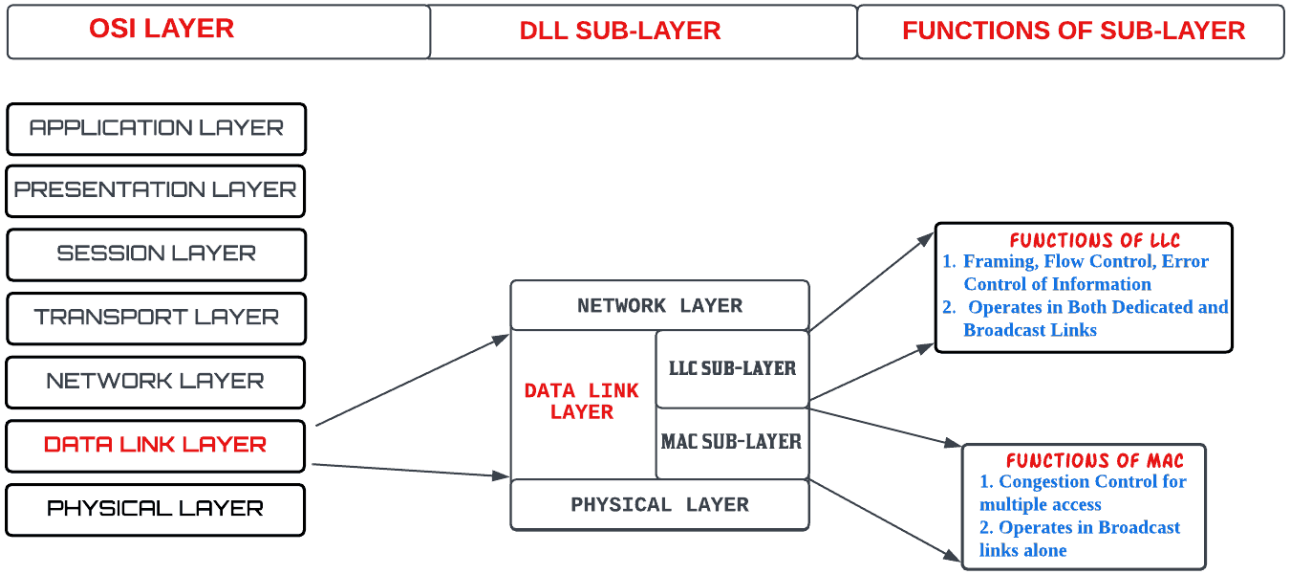
MAC objective

High utilization + fairness + acceptable delay.

- **MAC is a coordination problem at the link layer, especially important on LANs and wireless networks.**

MAC Layer Context & Allocation

- The **MAC Sublayer** is the lower sublayer of the Data Link Layer (OSI Model), responsible for deciding who sends next on a multi-access link.
- Responsible for deciding who sends next on a multi-access link
- An important part of the link layer, especially for LANs



The Channel Allocation Problem

- **Static Allocation (FDM, TDM):** Divides bandwidth into fixed portions. Highly inefficient for bursty network traffic as unused slots are wasted.
- **Dynamic Allocation:** Gives the channel to a user only when they need it, vastly improving efficiency.
- **Key Assumptions:** Single channel, observable collisions, continuous or slotted time, and carrier sensing capabilities.

Assumption	Implication
Independent traffic	Often not a good model, but permits analysis
Single channel	No external way to coordinate senders
Observable collisions	Needed for reliability; mechanisms vary
Continuous or slotted time	Slotting may improve performance
Carrier sense	Can improve performance if available

Static allocation breaks under bursty traffic

A fixed share is predictable—but often idle

TDM example



Low load

Many reserved slots are empty.

Bursty load

A station may need more capacity briefly.

Dynamic alternative

Let active stations compete for unused capacity.

- **Static allocation is attractive when traffic is steady; dynamic allocation is better when demand is bursty.**

ALOHA Protocols

Foundational contention-based protocols allowing transmission whenever data is ready. Characterized by low delay under low load, but high collision probability.

CSMA (Carrier Sense Multiple Access)

Improves efficiency by "listening" to the channel before transmitting. Includes variants like 1-persistent, non-persistent, and p-persistent.

Collision-Free Protocols

Eliminates collisions entirely using reservation systems (Bitmap), token passing (Token Ring), or arbitration (Binary Countdown). Highly efficient under heavy load.

Wireless LAN Protocols

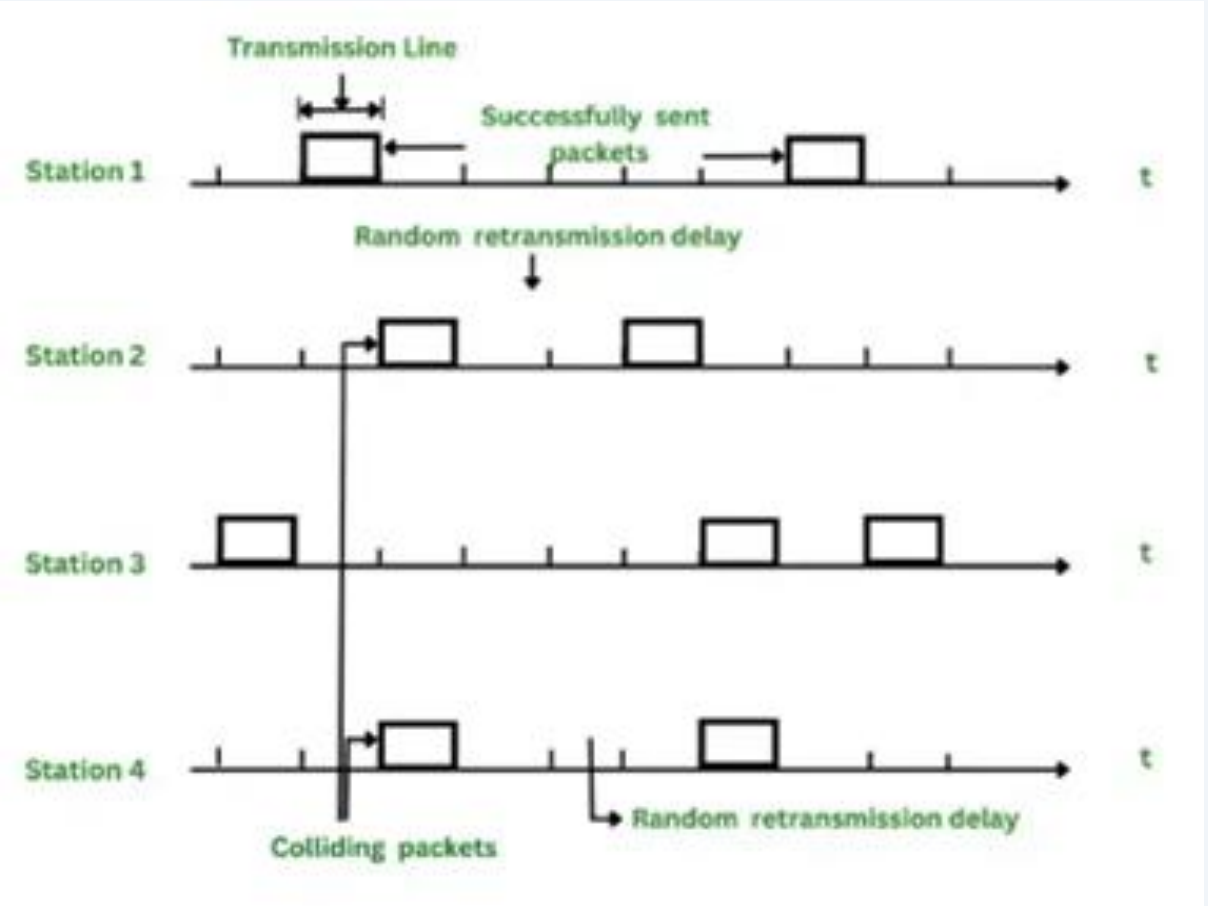
Adapted for environments where hidden and exposed terminals exist, primarily utilizing CSMA/CA (Collision Avoidance) instead of Collision Detection.

The ALOHA Protocol

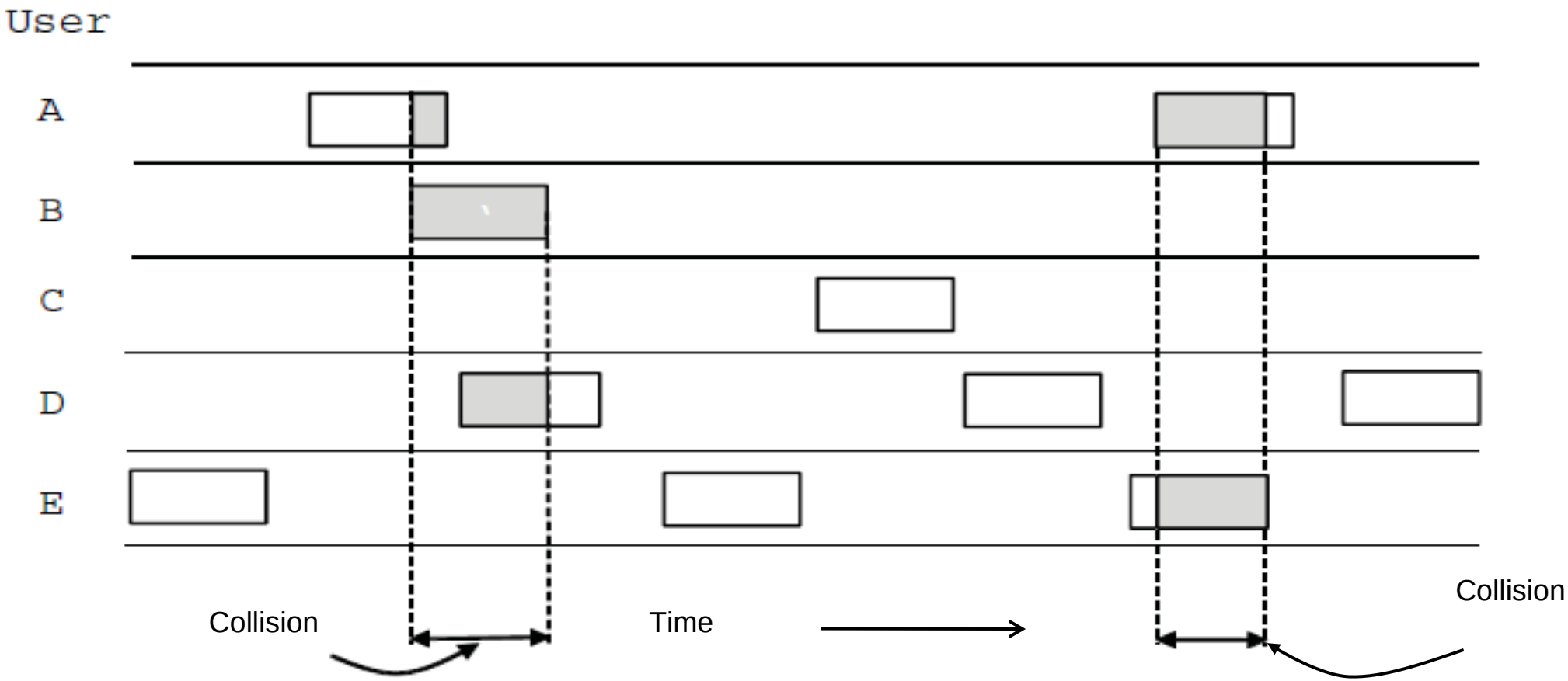
Pure vs. Slotted ALOHA

Pure ALOHA: Users transmit immediately. Collisions occur if transmissions overlap during a vulnerable period of twice the frame time (2t).

Slotted ALOHA: Time is divided into discrete slots. Transmissions must begin at slot boundaries, reducing the vulnerable period to t.



Pure ALOHA: transmit first, recover later

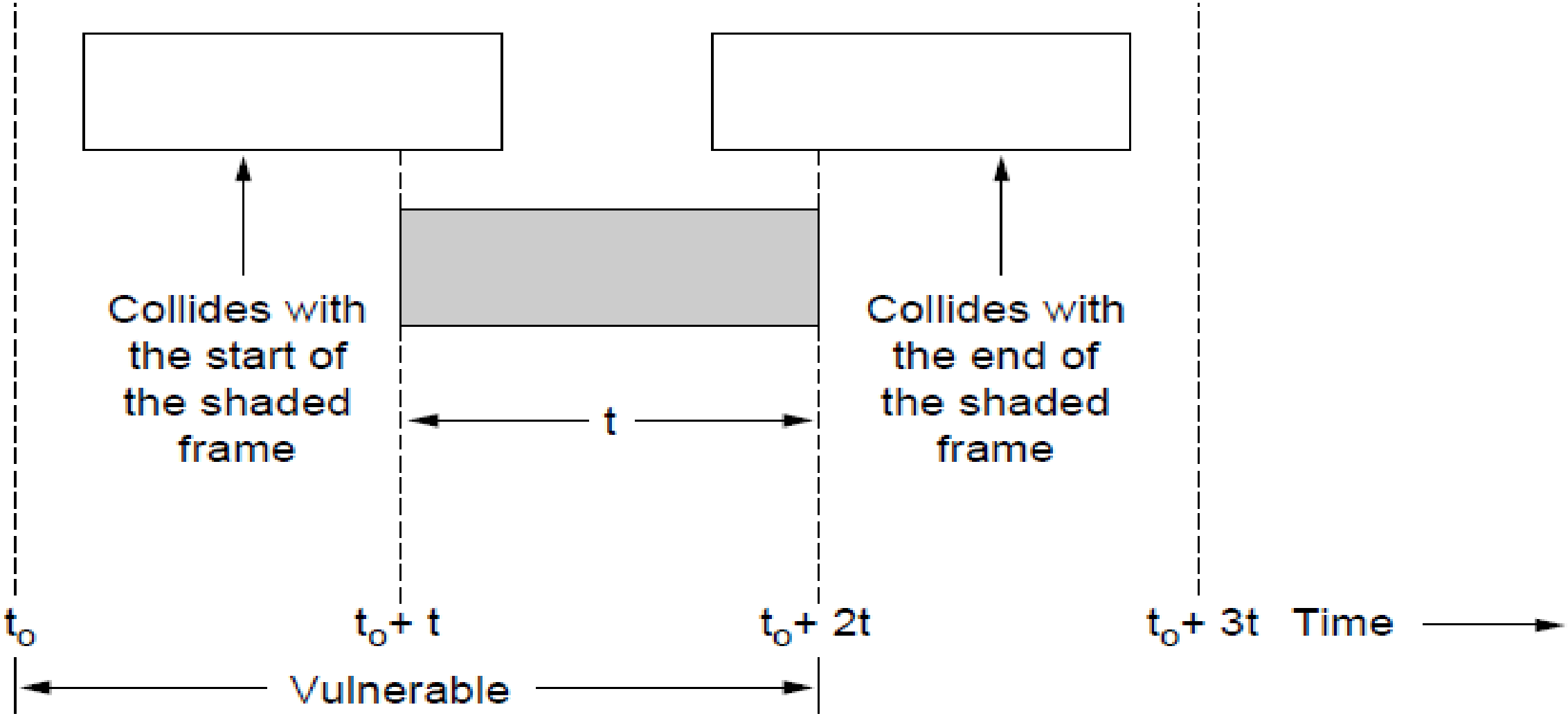


- **ALOHA is the baseline: every improvement tries to reduce the probability or cost of collisions.**

Why pure ALOHA wastes capacity

The vulnerable period explains the throughput limit

Suppose one frame takes T seconds to transmit.



Slotted ALOHA: one simple change

Start transmissions only at slot boundaries



A collision now requires two or more stations to choose the same slot.

Benefit

Vulnerable period is cut roughly in half.

Cost

All stations need slot synchronization.

Maximum throughput

≈ 37% in the classic model.

TEACHING POINT • Slotted ALOHA demonstrates a recurring design pattern: add coordination to recover efficiency.

Efficiency of ALOHA

- frametime – time needed to transmit standard sized frame
- N – New frames per frame time by poison distribution
- If N>1, nearly every frame will suffer a collision. 0<N<1 expected for reasonable throughput
- G – mean number of frames/frametime including retransmissions. Clearly, G ≥ N.
- At low load (i.e.,N~0), there will be few collisions, hence few retransmissions, so G≈N
- At high load G>N
- Throughput S is G x probability P_0 of successful transmission
- $S = G P_0$
- The probability that k frames are generated during a given frame time in which G frame were expected, is given by the Poisson distribution $P_r[k] = \frac{G^k . e^{-G}}{K!}$
- So probability of no frame is e^{-G}
- In an interval two frame times long, the mean number of frames generated is 2G. The probability of no frames being initiated during the entire vulnerable period is thus given by $P_0=e^{-2G}$. Using $S=GP_0$, we get $S=Ge^{-2G}$
- The maximum throughput occurs at G=0.5, with S=1/2e, which is about 0.184

ALOHA comparison

Same shared medium; different collision exposure

Pure ALOHA

Transmit at any time.

Slotted ALOHA

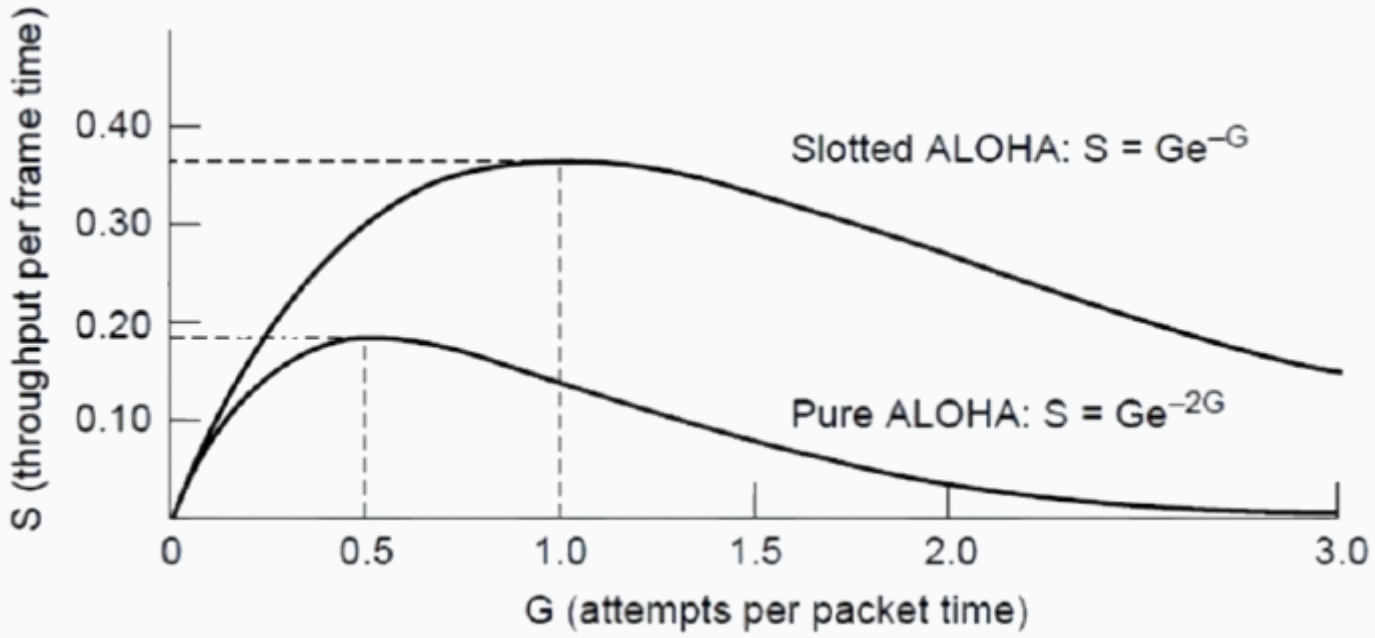
Transmit only at slot boundaries.

	Pure ALOHA	Slotted ALOHA
Start rule	Any time	Slot boundary
Vulnerable period	2T	T
Classic max throughput	≈ 18.4%	≈ 36.8%
Synchronization	Not required	Required

TEACHING POINT • Do not memorize the percentages alone: understand why slotting cuts the vulnerable period.

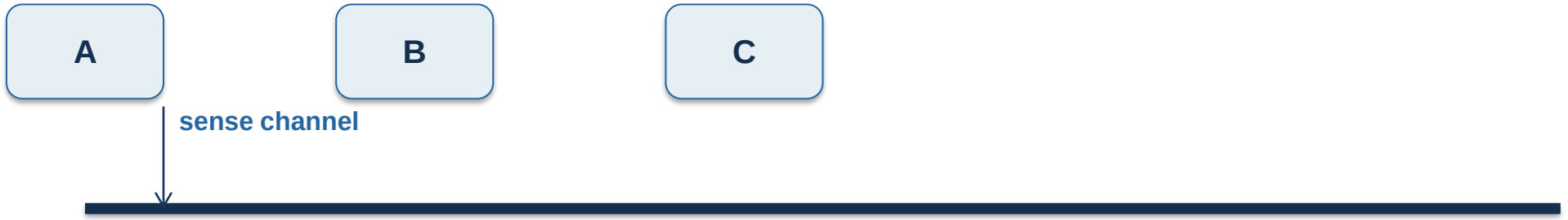
Efficiency of ALOHA

- Slotted ALOHA is twice as efficient as pure ALOHA
- Low load wastes slots, high loads causes collisions
- Efficiency up to 1/e (37%) for random traffic models



CSMA: listen before speaking

Carrier Sense Multiple Access reduces avoidable collisions



Idle → transmit

A station can send when it sees the channel idle.

Busy → defer

A station waits instead of immediately colliding.

But not perfect

Propagation delay means two stations may sense idle simultaneously.

- **CSMA attacks collisions before transmission; it does not eliminate collisions.**

CSMA persistence strategies

What should a station do after finding the channel busy?

1-persistent

Keep sensing; transmit immediately when idle.

Nonpersistent

If busy, wait a random time before sensing again.

p-persistent

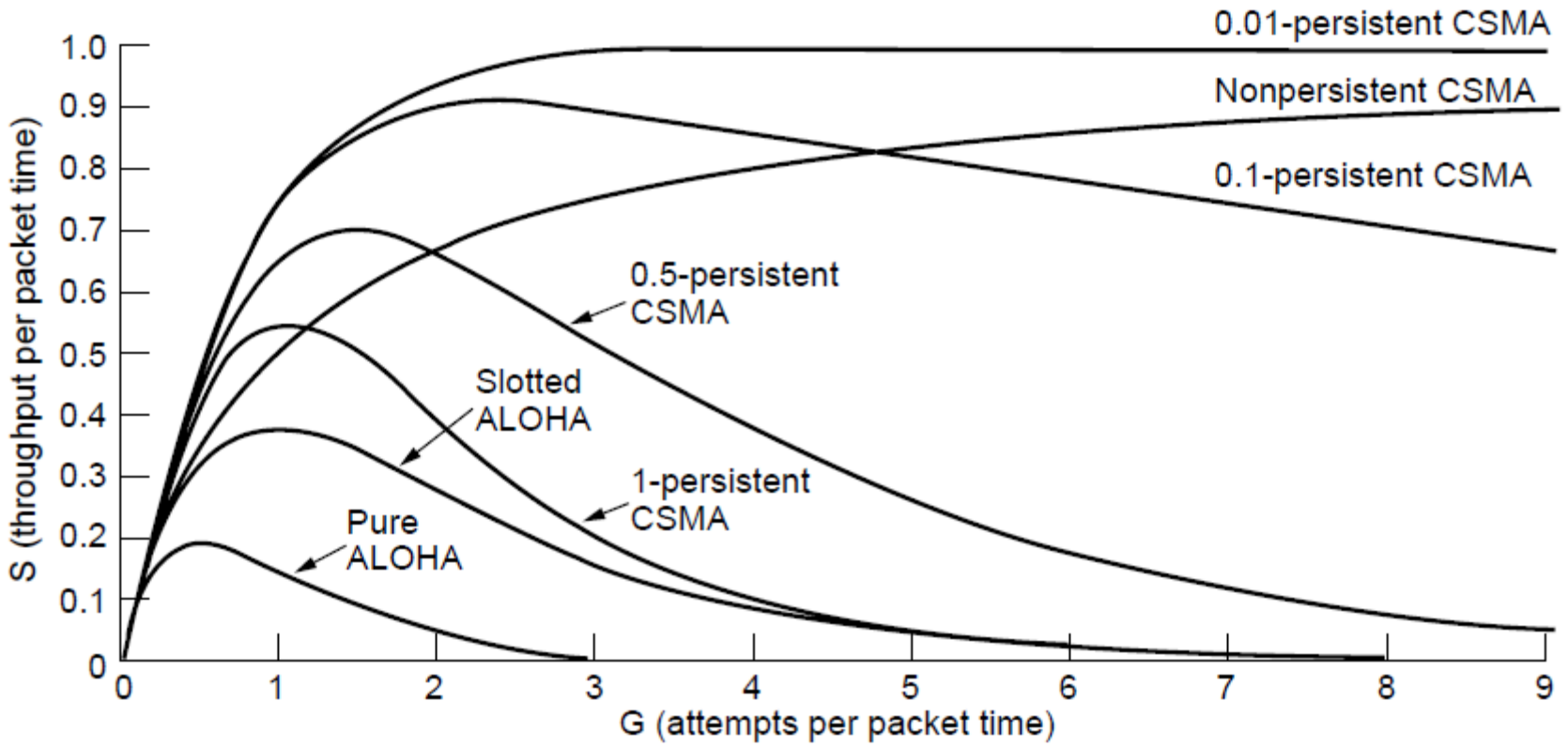
When idle, transmit with probability p; otherwise defer one slot.

Aggressive persistence helps under light load, but many eager stations can collide when the channel becomes free.

- **Under heavy contention, being slightly less eager can improve overall throughput.**

CSMA persistence efficiency

CSMA outperforms ALOHA, and being less persistent is better under high load



Why CSMA can still collide

Propagation delay creates a “shared illusion”



A begins transmitting

D has not heard A yet → also transmits

Propagation delay

Signals need time to travel across the medium.

Collision window

Two stations can decide to transmit before hearing each other.

Design response

Detect collisions early where possible.

- **Distance matters: longer propagation delay means more uncertainty about other stations' activity.**

CSMA/CD: detect and abort

Classic shared Ethernet shortened collision waste



Instead of completing a doomed frame, the station stops as soon as it detects another signal.

Practical on wired shared media because a transmitter can monitor the medium while sending.

Problematic in wireless: the local transmitted signal can overwhelm the received signal.

- **CSMA/CD makes collisions cheaper; modern switched Ethernet largely removes the need for collision detection.**

Binary Exponential Backoff

After a collision, contenders spread themselves out



More collisions → larger random window → lower probability that the same stations immediately collide again.

Benefit

Adaptive under load.

Trade-off

Delay grows after repeated collisions.

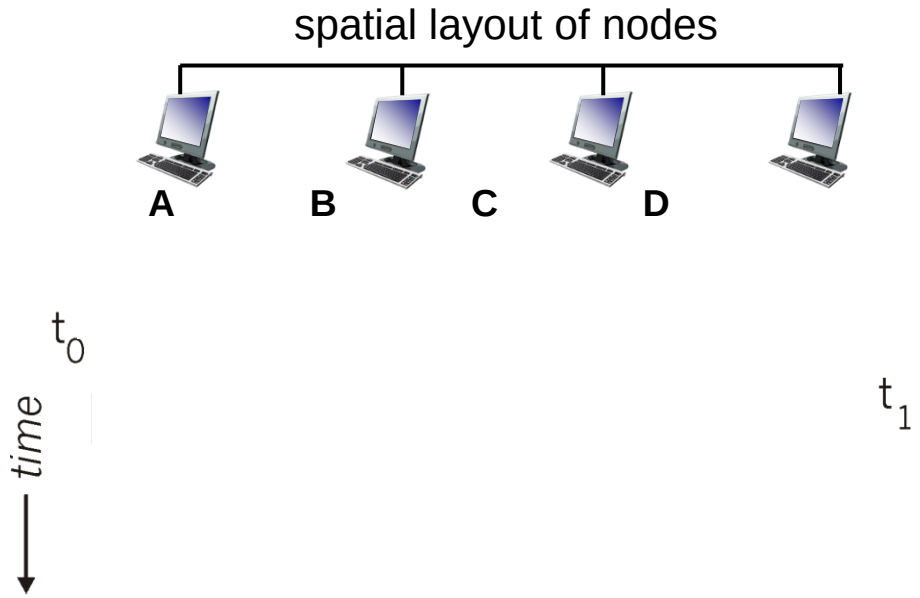
Key idea

Collision history becomes feedback.

- **Backoff is feedback: collisions are evidence that contention is high.**

CSMA Collisions

- Collisions can still occur: propagation delay means two nodes may not hear each other's transmission
- Collision: entire packet transmission time wasted
- Distance & propagation delay play role in in determining collision probability



CSMA Collisions Detection

- A collision is detected by the transmitting station while it is transmitting
- If the signal it observes on the cable is different from what it is putting onto the cable, it concludes that another station is transmitting too → collision.
- Shared Ethernet cable

A ——————————————————————— B

→ A's signal ← B's signal

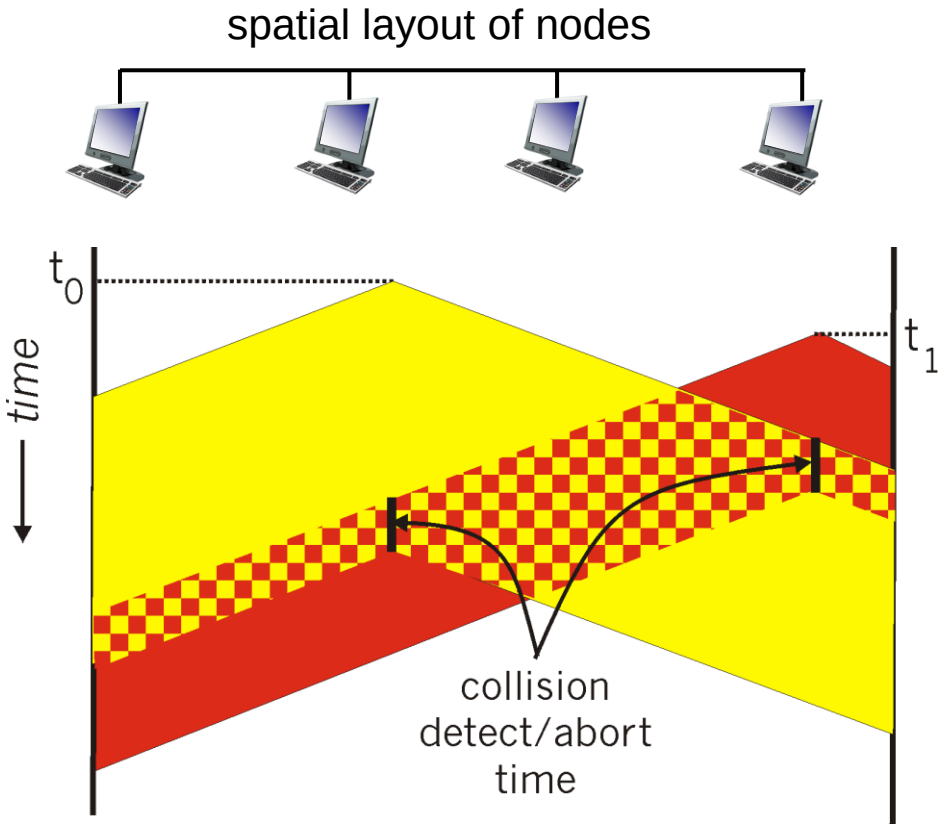
- At some point, their electrical signals overlap:

A sends: + + + + + + + + +

B sends: - - - - - - - - -

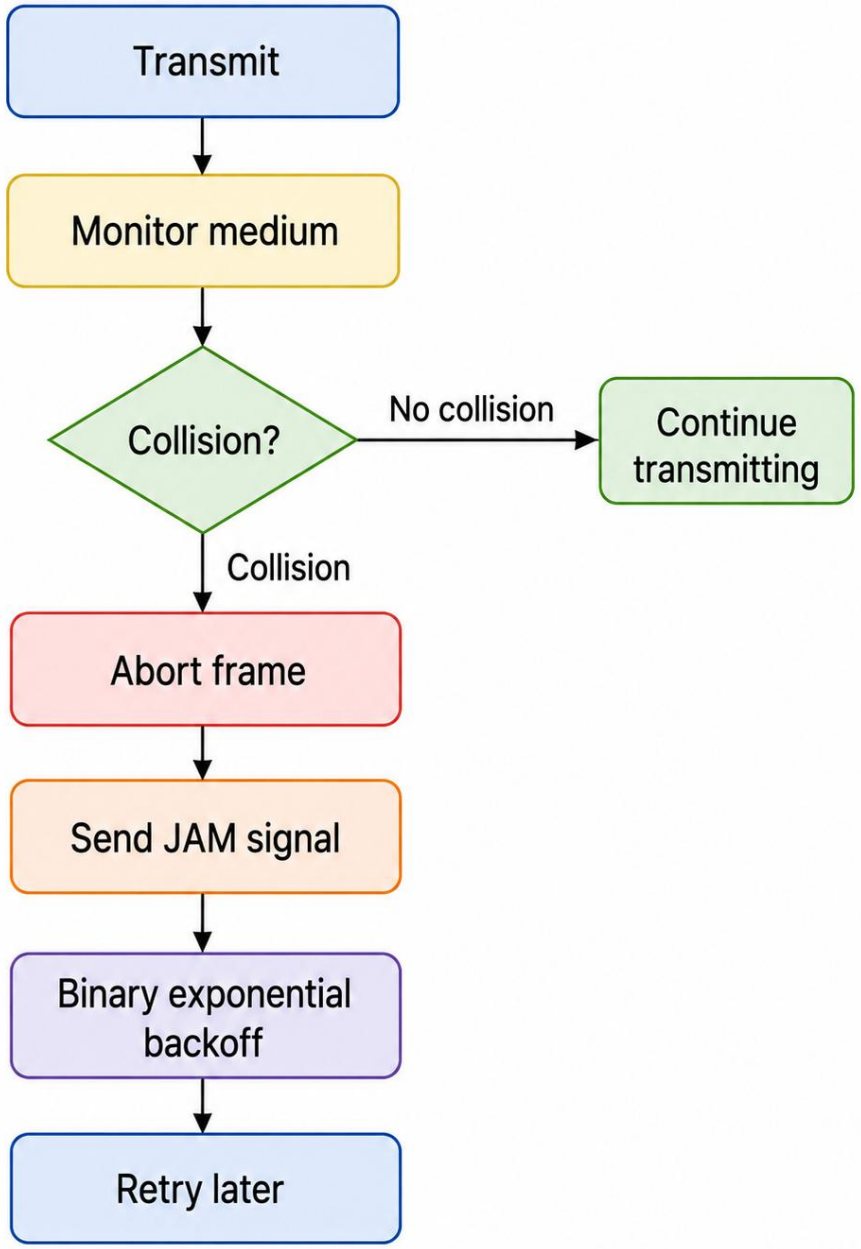
On cable: + - + - + - + - + -

↑ corrupted signal



CSMA Collisions Detection

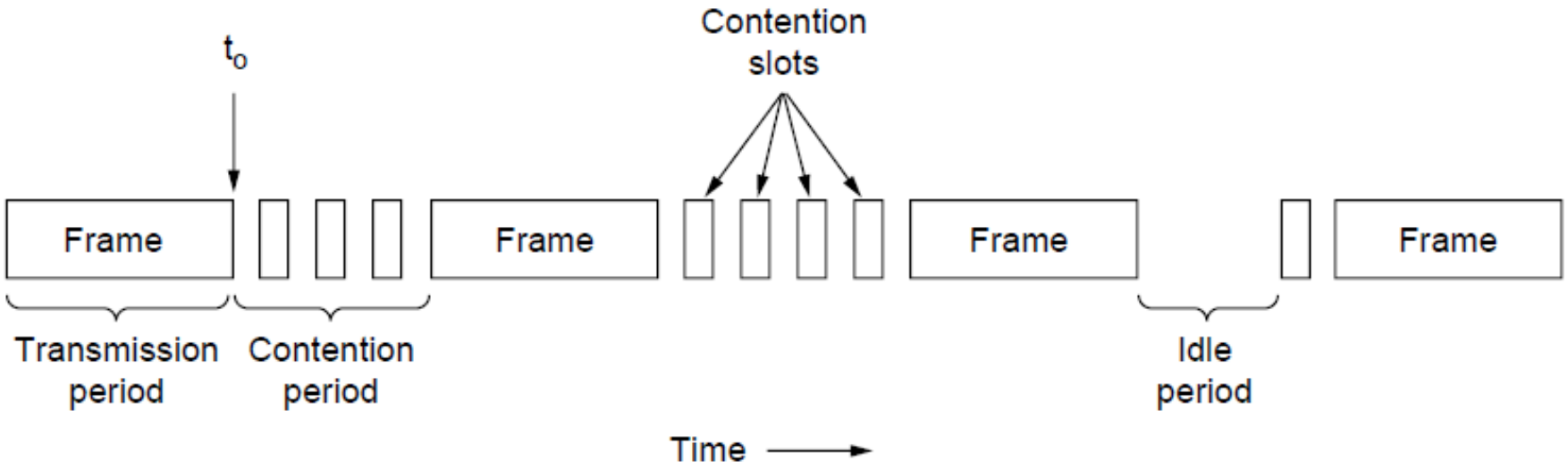
- While transmitting, A's Ethernet interface monitors the voltage/signal actually present on the cable.
- What A sends → Compare ← What exists on cable
- If do not match → Collision
- It is therefore not that A receives a special "collision packet.“
- Instead, the physical-layer transceiver detects abnormal electrical activity / a mismatch between transmitted and observed signals.
- A JAM signal is sent to ensure that all stations on the shared medium become aware that a collision has occurred and that the current frame should be discarded.
- A station must transmit long enough to guarantee it will hear a collision if one occurs (minimum frame size requirement based on 2x propagation delay).



This forms the foundation of classic wired Ethernet.

CSMA Collisions Detection

- Transmissions are aborted when a collision is detected leading to medium being free
- Contention begins and winner transmits again with possibility of collision
- Transmission period → When a station has acquired the medium without detecting collision
- Contention period → Fight for the medium, till some one acquires the medium. Might have multiple contention slots
- Idle period → None of the station is ready to transmit



Collision Free Protocols / Reservation-based Protocols

- Collisions adversely affect the system performance as it leads to wastage of line bandwidth and transmission efforts
- If medium can be reserved for prospective transmission by efficiently scheduling the transmissions by different stations, collisions can be removed/reduced
- Collision-free protocols avoid collisions entirely
- Senders must know when it is their turn to send
- Few Assumptions
  - There are exactly N stations, each programmed with a unique address from 0 to N−1.
  - Propagation delay between the stations is negligible

Collision-Free Bitmap Protocols

The Basic Bitmap Protocol

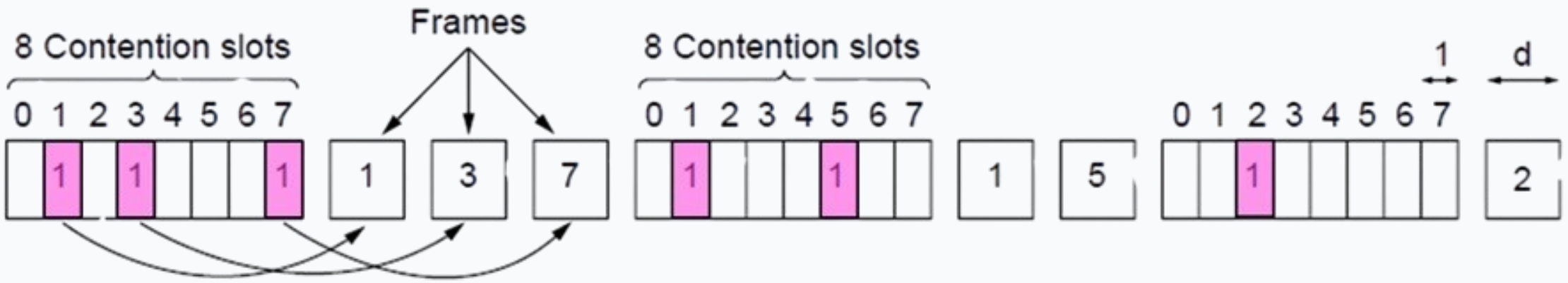
- The protocol divides communication into two phases:
  - Contention / reservation phase — stations announce whether they want to transmit.
  - Data transmission phase — stations that announced themselves transmit one at a time in station-number order.
- The bitmap determines who gets to transmit before the actual data transmission begins.
- How bit map is created?
  - The protocol provides N bit reservation slots, one corresponding to each station.



  - The reservation bitmap is: 0 1 0 1 0 0 1 0
- How bit map is created?
  - Stations transmit according to their station number, only those stations which have announced 1 during their slot
  - There is no competition during this phase. Hence no collision.

Collision-Free Bitmap Protocols

The Basic Bitmap Protocol



Collision-Free Bitmap Protocols

The Basic Bitmap Protocol Performance

- Measure the time in units of the contention bit slot, with data frames consisting of d time units
- The network can be highly loaded i.e. most stations are willing to transmit or lightly loaded when very few (about 1 station) is willing to transmit
- Under conditions of low load
  - A lower numbered station has to wait $N + \frac{N}{2}$ slots
  - Higher numbered station has to wait for $\frac{N}{2}$ slots
  - The mean delay for all station is N slot
  - Channel efficiency: The overhead per frame is N bits and the amount of data is d bits, for an efficiency of $\frac{d}{(d+N)}$
- Under conditions of high load
  - The mean delay for a frame is $(N-1)d + N$
  - Channel efficiency: The overhead per frame is 1 bits for an efficiency of $\frac{d}{(d+1)}$

Collision-Free Bitmap Protocols

The Basic Bitmap Protocol Issues

- Stations' access to the network is unfair: That is, if station i and station j both want to transmit, and i < j, then station i is always first to transmit.
- Low numbered stations have to wait longer than high numbered stations for the reservation to complete.
- Efficiency: at low load, the protocol efficiency is low.

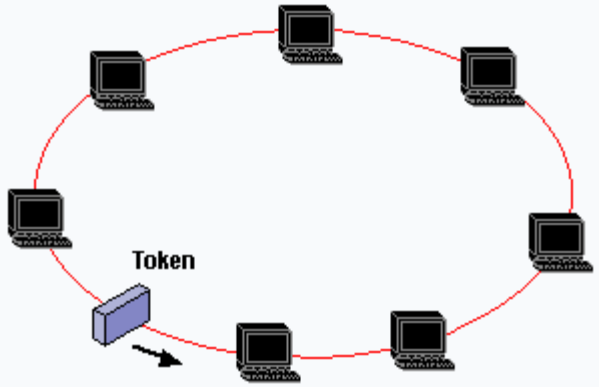
The Basic Bitmap Protocol – Comparison with CSMA/CD

CSMA/CD	Basic Bitmap
Stations compete to transmit	Stations reserve transmission opportunity
Collisions can occur	Data collisions are avoided
Collision detection required	No collision detection needed
Random access	Controlled access
Good when load is relatively light	Good when load is high
Retransmission after collision	No collision-based retransmission

Collision-Free Protocols – Token Ring

The Token Ring Network

- In a Token Ring, all stations are logically arranged in a ring, and a special control frame called a token continuously circulates around the ring.
- How it avoids collisions?
    1. Only the station holding the token is allowed to transmit.
    2. If A has data, it waits for the token.
    3. A captures the token and transmits its frame.
    4. While A is transmitting, no other station can transmit.
    5. After completing its transmission, A releases the token.
    6. The token moves to the next station..



> There is only one token, and possession of the token gives exclusive permission to transmit ensuring collision-free operation

The Token Ring Network – Comparison with CSMA/CD

- CSMA/CD: Transmit → collision may occur → detect → abort → retry
- Token Ring: Wait for token → transmit → release token

Collision-Free Protocols – Binary Countdown

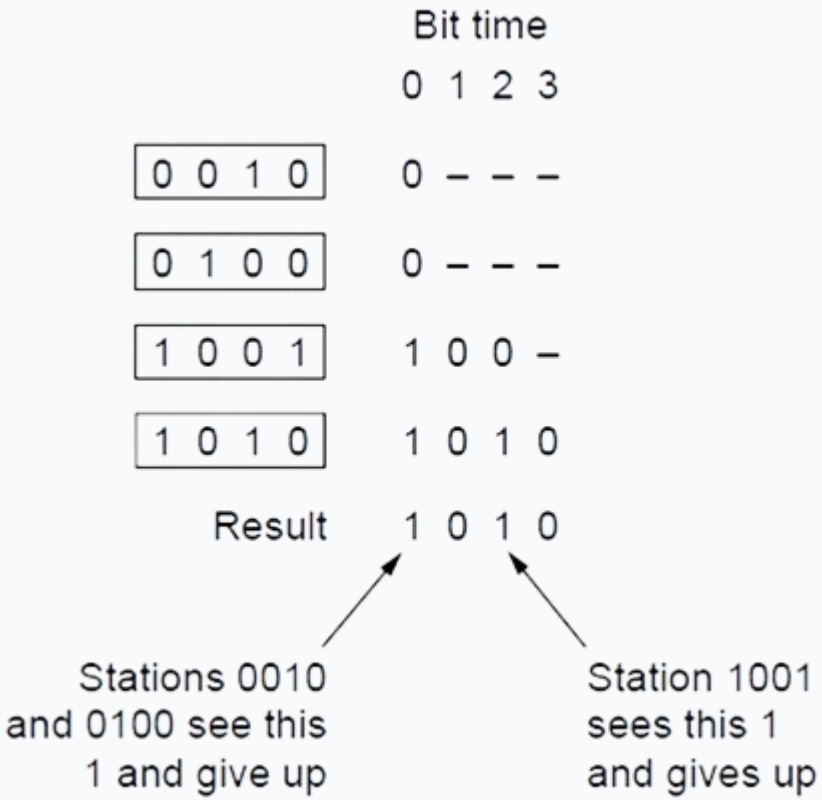
The Binary Countdown Protocol

- Like basic bitmap, binary countdown also has two phases- Reservation phase and Transmission phase.
- Binary countdown improves on the bitmap protocol by reducing the reservation overhead from n to log n
- Reservation
    1. All contenders broadcast their address bit by bit, starting from the most significant bit (MSB).
    2. At each bit position, stations effectively perform a wired-OR.
    3. If a station transmitting 0 detects that another station has transmitted 1, it will give up in the current reservation cycle.
    4. At the end of log n rounds, only one station will remain and that station transmits.
- Binary countdown eliminates collisions, but it requires a reservation/countdown phase.
- Binary countdown allows stations to compete bit-by-bit using their addresses; lower-priority contenders withdraw as soon as a higher-priority bit appears, leaving exactly one station to transmit without collision.

> It is called binary countdown because the stations essentially count down through their binary addresses, starting with MSB.

Collision-Free Protocols – Binary Countdown

The Binary Countdown Protocol



Collision-Free Protocols – Binary Countdown

The Binary Countdown Protocol - Performance

- The biggest advantage is that collisions are avoided completely.
- Contention overhead:
  - Every transmission opportunity requires a countdown phase.
  - If addresses have k bits, the competition requires approximately k bit times.
  - For N stations, $\log_2 N$ bit times will be required to resolve contention.
- Assume each frame of d bits. For transmission of a frame, the overhead is logN bit times, leading to efficiency of $\frac{d}{(d+\log_2 N)}$

Contention-based and Collision-Free Protocols – Summary

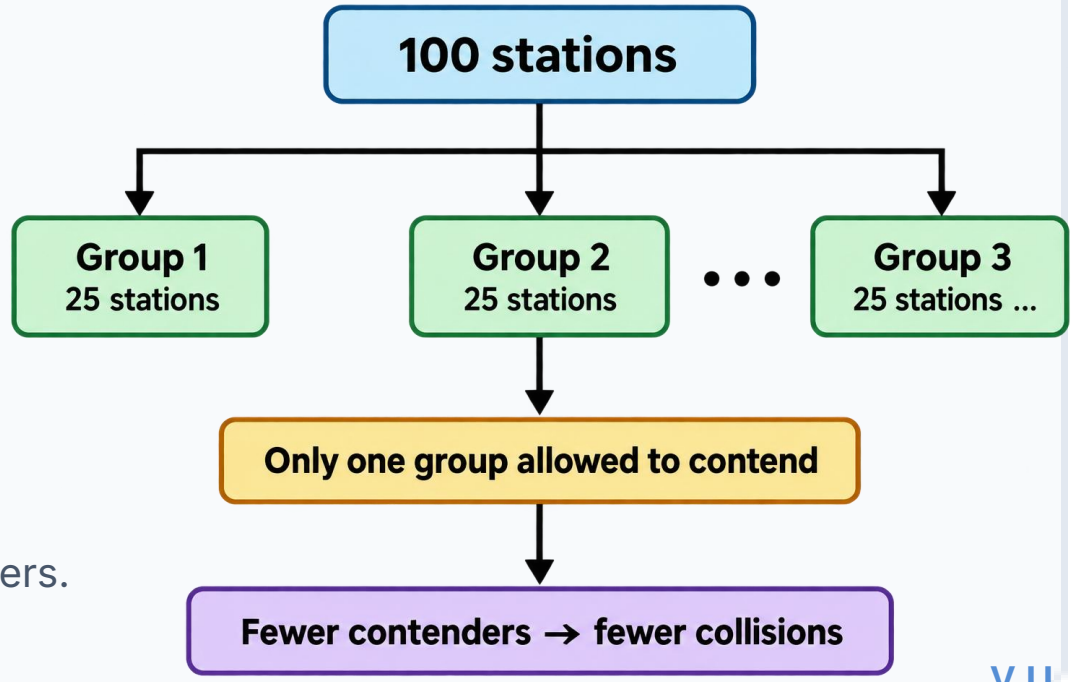
The Binary Countdown Protocol - Performance

- It is concluded that
  - Contention-based systems perform better at lower load
  - Reservation-based protocols perform better at higher load.
- Is it possible to design an adaptive solution which works like contention system at lower load and like reservation system when load increases?

Limited Contention Protocols – A balance between two extremes

The Limited Contention Protocol

- Limited contention protocols try to get the best of both worlds:
  - **Contention-based protocols** such as CSMA are efficient when few stations are active.
  - **Collision-free protocols** such as bitmap and binary countdown are efficient when many stations are active.
- The problem is that one protocol does not perform optimally at all traffic loads.
- Limited contention protocols attempt to reduce the number of stations competing for the channel at any one time.
- Basic Concept:
- With normal CSMA → N stations contend → Collisions → Backoff
- Limited contention protocol divides stations into groups.
- Permit only selected group of stations to contend, thus limiting the contention
- A hierarchy of groups will enable the protocol to control contention at different levels
- Limited contention tries to maintain a manageable number of contenders.



Limited Contention Protocols – A balance between two extremes

Probability of successful transmission

- Suppose k stations are currently competing, and each station independently decides to transmit in a particular slot with probability p (p-persistent)
- For a successful transmission, exactly one station must transmit.
  - Choose which station transmits → k possibilities
  - That station transmits → probability p
  - All other k–1 stations remain silent → probability $(1-p)^{(k-1)}$
  - $P_{success} = k\,p\,(1-p)^{(k-1)}$
- To find optimal value of p, differentiate w.r.t. p, equate it 0 and solve for p.
- Differentiate with respect to p: $\frac{dP}{dp} = k(1-p)^{k-1} - kp(k-1)(1-p)^{k-2}$
- For maximum success probability, set the derivative to zero: $k(1-p)^{k-1} - kp(k-1)(1-p)^{k-2} = 0$
- Take common terms out: $k(1-p)^{k-2}[(1-p) - p(k-1)] = 0$
- So, $(1-p) - p(k-1) = 0, \qquad 1 - kp = 0, \quad p = \frac{1}{k}$
- The probability of success is maximized when p =1/k
- Substituting gives Probability of success (with optimal p) = $[\frac{k-1}{k}]^{(k-1)}$

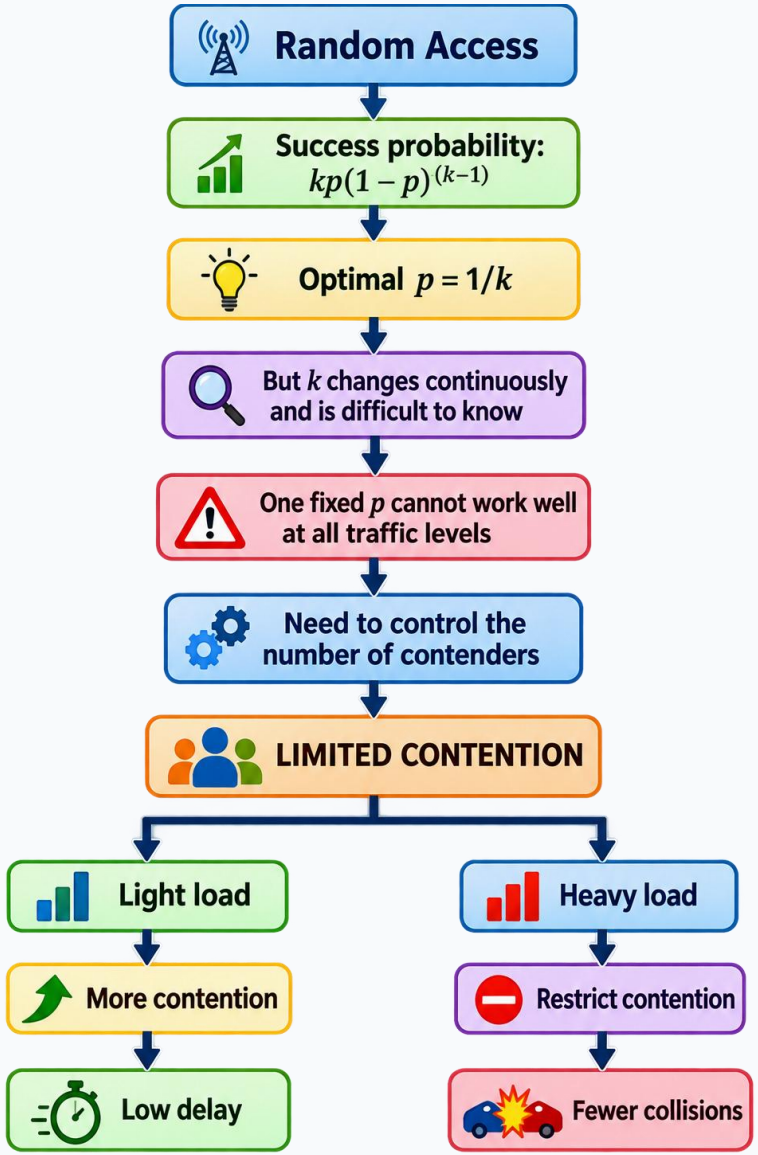
Limited Contention Protocols – A balance between two extremes

Probability of successful transmission

- Substituting gives Probability of success (with optimal p) = $[\frac{k-1}{k}]^{(k-1)}$
- As the number of competing stations becomes very large, $[\frac{k-1}{k}]^{(k-1)} \rightarrow \frac{1}{e}$
- The max $P_{success,max} \approx 0.368$, even with the optimal value of p
- So even with the optimal transmission probability, when there are many competing stations, the maximum probability that a slot contains exactly one successful transmission is only about 36.8%.
- The optimal value of p depends on number of ready stations which is difficult to determine.
- A transmission probability that works well for light traffic can perform terribly under heavy traffic.
- The key idea is: Don't allow every active station to contend at the same time. Limit the number of contenders.

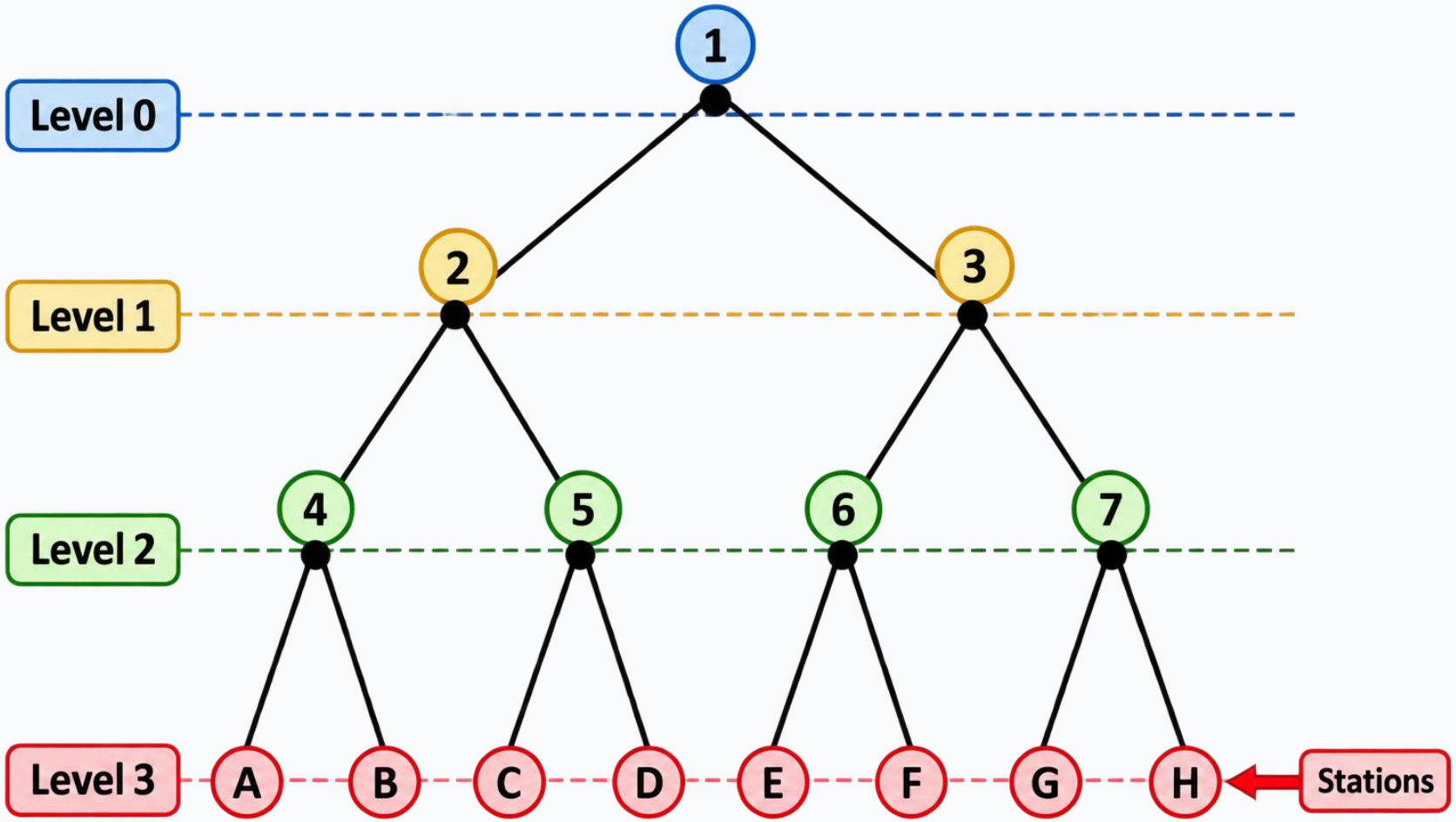
Limited Contention Protocols – A balance between two extremes

The evolution



Limited Contention Protocols – A balance between two extremes

The Limited Contention Protocol



The Evolution of Ethernet

Bus Ethernet

- Thick Ethernet - Thick coaxial cable with segment length of 500 m with each segment can have 100 PCs
- Thin Ethernet - Thin and flexible coaxial cable with segment length of 185 m with each segment can have 30 PCs
- Repeaters can be used to extend the length of Ethernet beyond the max segment length

.

From Hubs to Switches

- Early Ethernet used **Hubs**, which wired all lines into a single, massive collision domain (CSMA/CD domain). As networks grew, efficiency plummeted.
- Modern networks use **Switches**. A switch isolates each port into a separate collision domain. With full-duplex lines, CSMA/CD is no longer required, allowing simultaneous sending and receiving without collisions.

Fast, Gigabit, and Beyond

The need for speed drove rapid standard enhancements over twisted pair (UTP) and fiber optics.

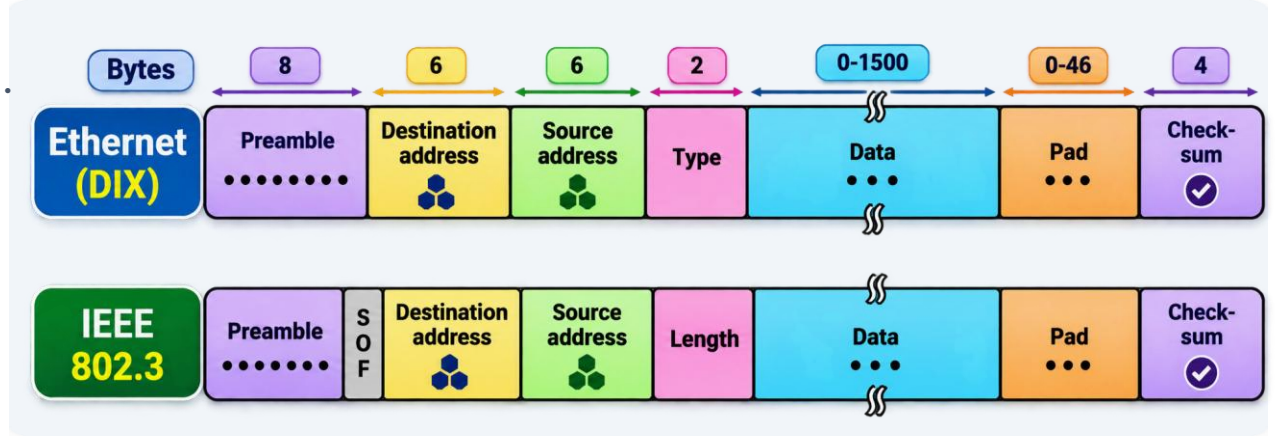
- **Fast Ethernet:** Increased bit rate from 10 Mbps to 100 Mbps (e.g., 100Base-TX).
- **Gigabit Ethernet:** 1000 Mbps throughput. Point-to-point switched topology became the garden variety
- **10-Gigabit & Up:** Backbone deployments expanding to 40/100 Gbps standards for data centers.

Classic Ethernet (IEEE 802.3)

The Ethernet Frame

Classic Ethernet uses 1-persistent CSMA/CD over a shared medium (like a coaxial cable), operating at 10 Mbps with Manchester encoding.

- **Preamble:** 8 bytes used for clock synchronization. 101010... pattern is used, with the last 2 bits set to 11. Using Manchester encoding, it produces a square wave
- **Addresses**: 6-byte MAC addresses (globally unique).
- **Type/Length**: type indicates the kind of content carried by the frame, like 0x0800 means IPv4. If the number is less than 0x600 (1536), it is interpreted as length; otherwise, as type, to ensure compatibility between both versions.
- **Data Payload**: Up to 1500 bytes.
- **Pad:** 0 to 46 bytes to make the total length 64 bytes
- **Checksum**: To protect the frame against errors

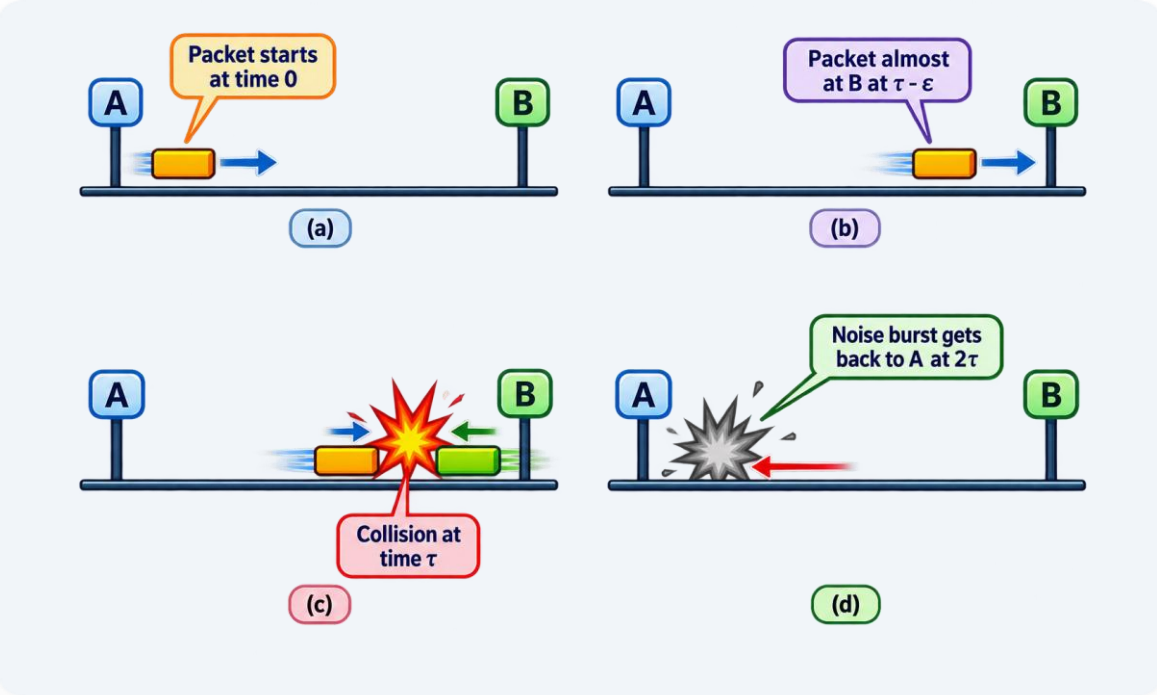


Classic Ethernet (IEEE 802.3)

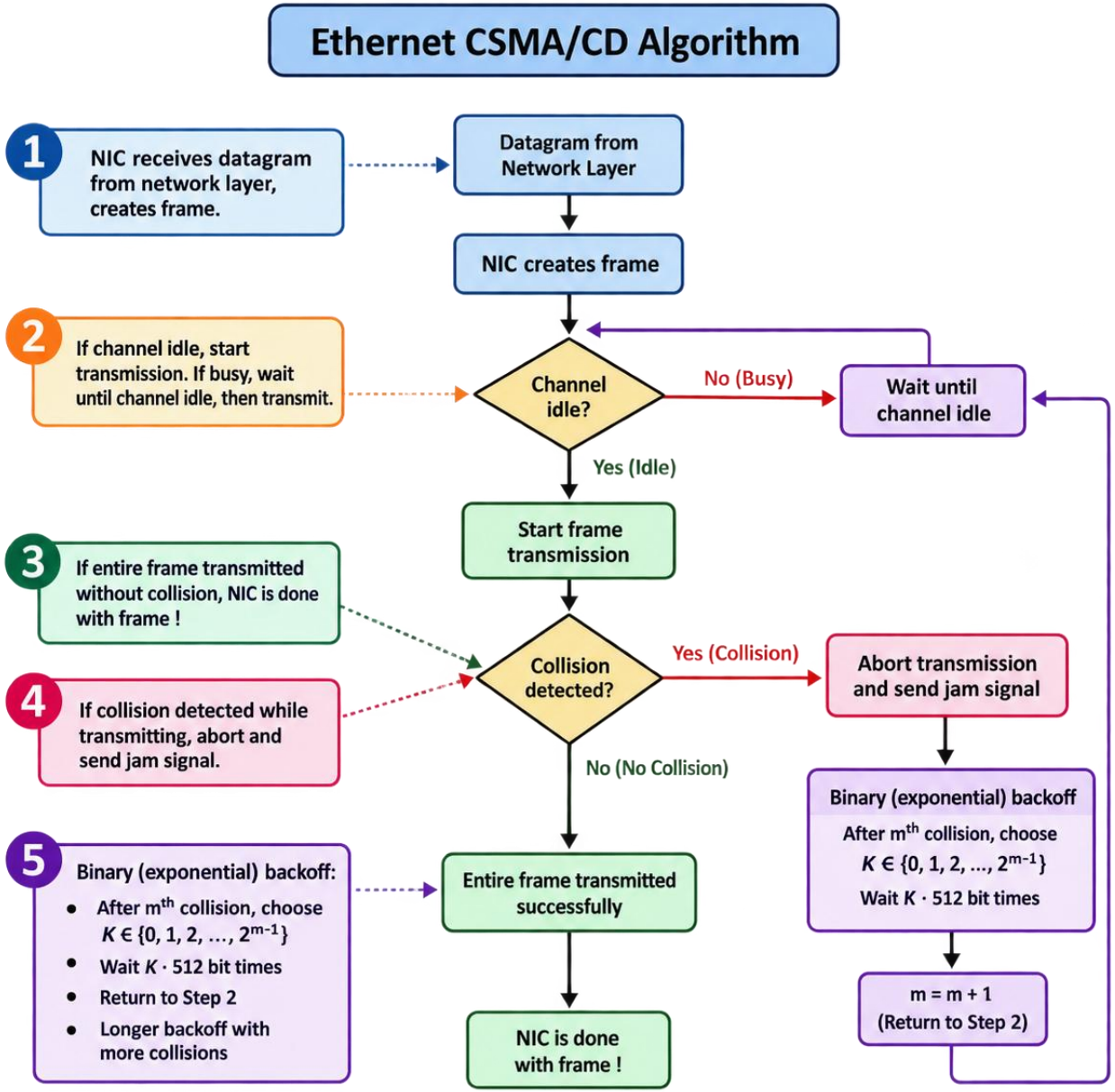
Collision Detection

Collisions can occur and take as long as 2τ to detect

- τ is the time it takes to propagate over the Ethernet
- Leads to minimum packet size for reliable detection



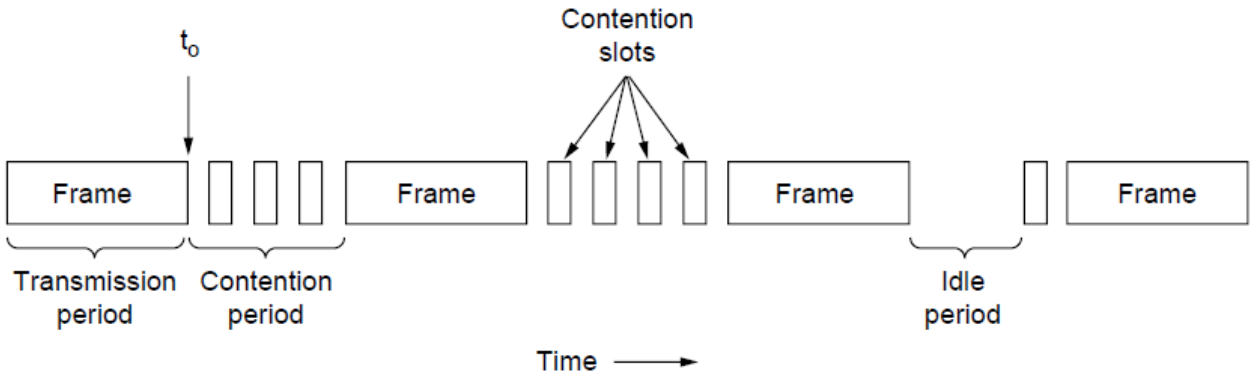
Classic Ethernet (IEEE 802.3)



Classic Ethernet (IEEE 802.3)

Performance

Remember



To analyse the performance of Ethernet

- Try to find mean duration of contention interval OR
- Mean number of contention slots in a contention interval

Classic Ethernet (IEEE 802.3)

Performance

- Assume k nodes are contending in the contention period
- Assume each node transmits with a fixed probability p in any slot
- What is the probability ('A') that some station acquires the channel in a slot (for successful transmission)?
- $A = p(1-p)^{(k-1)} + p(1-p)^{(k-1)} \ldots\ldots k\ times$
- $A = k\ p\ (1-p)^{(k-1)}$
- A is maximized when $p = {}^{1}/_{k}$, with $A \Rightarrow {}^{1}/_{e}$ as k tends to infinity.
- The probability (P_j) that a contention interval has exactly j slots?
  - After the collision, the least size of back-off interval chosen by some node is j slots or, what is the probability that some station transmits successfully only at j^{th} slot and not in previous j-1 slots?
  - p(Not transmitting in 1st slot)= (1-A)
  - p(Not transmitting in 1st slot and 2nd slot)= (1-A)(1-A)
  - p(Not transmitting in j-1 slots)= $(1-A)^{(j-1)}$
  - $P_j = A\ (1-A)^{(j-1)}$
- Over a long period of time, the mean number of slots per contention interval is given by $\sum_{j=0}^{\infty}(j\ * A\ (1-A)^{(j-1)})$
- Assuming optimal p, the mean number of contention slots is never more than 1/A or e

Classic Ethernet (IEEE 802.3)

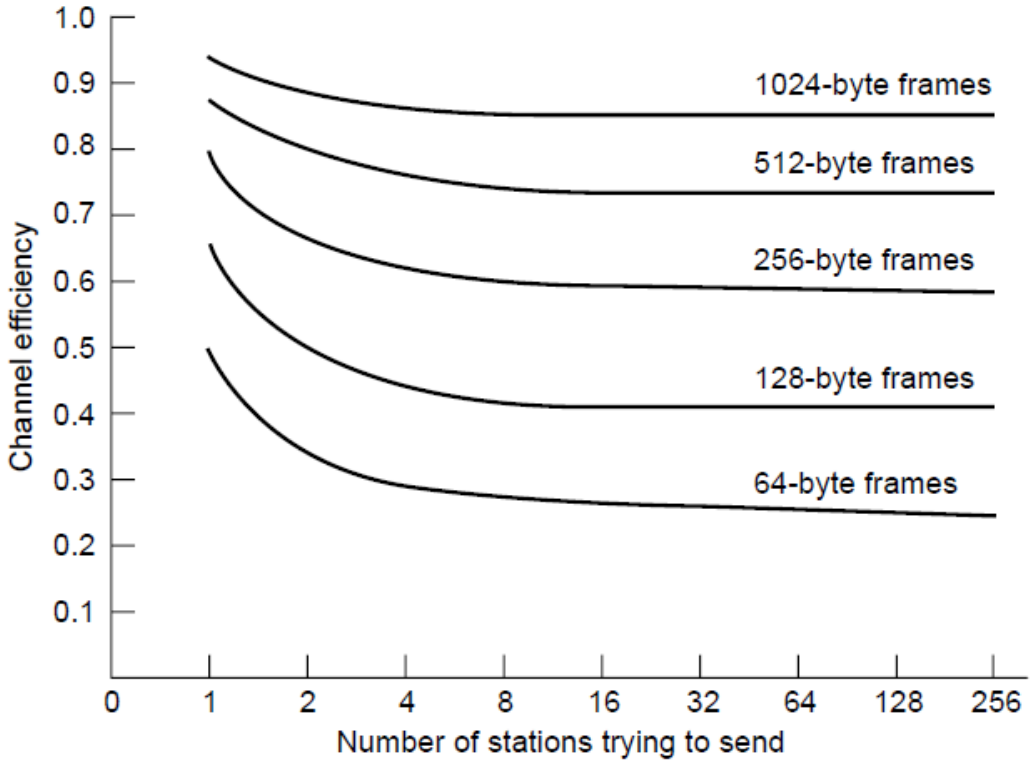
Performance

- Since each slot is 2*T_{prop},
- The mean length of contention interval is $T_{Contention} = 2 * {}^{T_{Prop}}/_{A}$
- Now, if T_{trans} is the time needed for transmitting the mean packet size, then
- $Channel\ Efficiency\ (w) =$ time required to transmit in absence of collision / time required to transmit in presence of collision
- $w = {}^{T_{trans}}/_{(Tt_{rans} + 2 * {}^{T_{prop}}/_{A})}$
- So, the longer the propagation time T_{prop} (cable length!), the longer the contention interval.
- The longer the contention interval, the lower the efficiency/throughput
- Assume the frame length is F, the network bandwidth, B, the cable length, L, and the speed of signal propagation, c
- $T_{trans} = {}^{F}/_{B}\ and\ T_{prop} = {}^{L}/_{c}$
- $w = {}^{T_{trans}}/_{(T_{trans} + {}^{2 * T_{prop}}/_{A})} = \frac{{}^{F}/_{B}}{({}^{F}/_{B} + {}^{2\ e\ L}/_{c})} = \frac{1}{(1 + {}^{2BLe}/_{cF})}$

Classic Ethernet (IEEE 802.3)

Performance

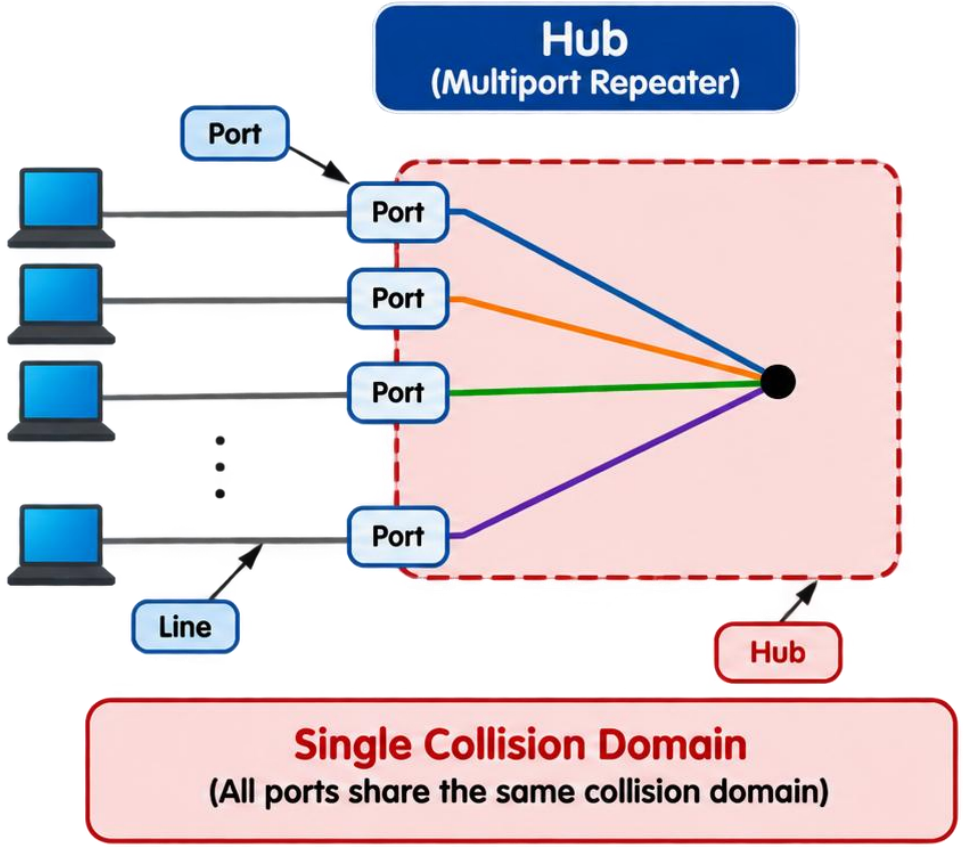
- Efficient for large frames, even with many senders
- Degrades for small frames (and long LANs)

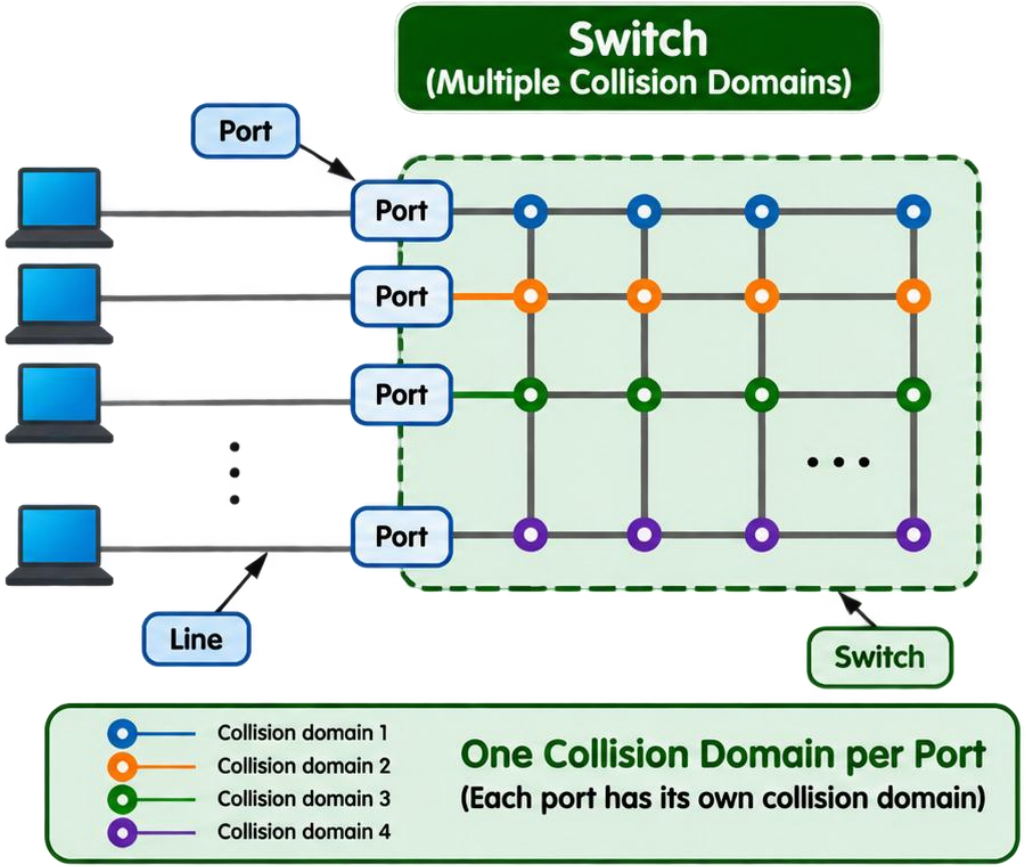


10 Mbps Ethernet,
64 byte min. frame

Switched Ethernet/FastEthernet

- Hubs wire all lines into a single CSMA/CD domain
- Switches isolate each port to a separate domain
  - Much greater throughput for multiple ports
  - No need for CSMA/CD with full-duplex lines





Switched Ethernet/FastEthernet

- Fast Ethernet extended Ethernet from 10 to 100 Mbps
- Twisted pair (with Cat 5) dominated the market

Name	Cable	Max. segment	Advantages
100Base-T4	Twisted pair	100 m	Uses category 3 UTP
100Base-TX	Twisted pair	100 m	Full duplex at 100 Mbps (Cat 5 UTP)
100Base-FX	Fiber optics	2000 m	Full duplex at 100 Mbps; long runs

- Gigabit Ethernet is commonly run over twisted pair

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

- 10 Gigabit Ethernet is being deployed where needed

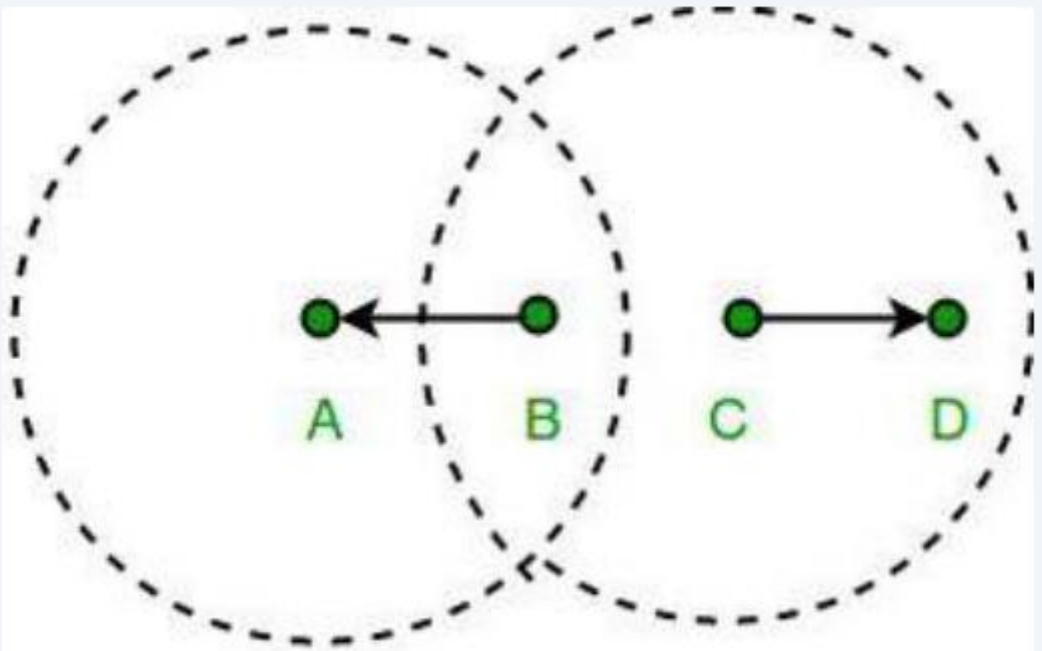
Name	Cable	Max. segment	Advantages
10GBase-SR	Fiber optics	Up to 300 m	Multimode fiber (0.85μ)
10GBase-LR	Fiber optics	10 km	Single-mode fiber (1.3μ)
10GBase-ER	Fiber optics	40 km	Single-mode fiber (1.5μ)
10GBase-CX4	4 Pairs of twinax	15 m	Twinaxial copper
10GBase-T	4 Pairs of UTP	100 m	Category 6a UTP

Wireless LAN Challenges

Hidden & Exposed Terminals

Unlike wired networks, radio nodes have limited, overlapping coverage regions, causing severe MAC challenges.

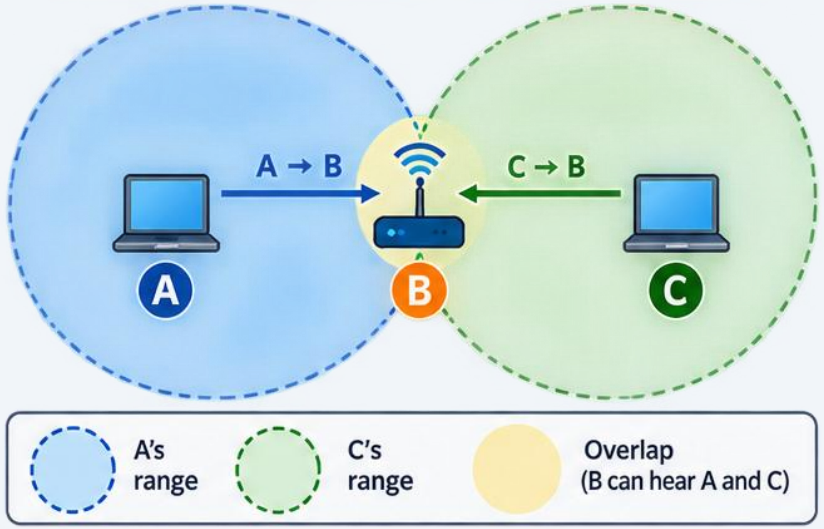
- **Hidden Terminal Problem:** Senders (A and C) cannot hear each other but transmit to the same receiver (B) simultaneously, causing unseen collisions.
- **Exposed Terminal Problem:** Senders mistakenly believe the channel is busy because they hear a nearby transmission, wasting bandwidth by deferring safe transmissions.
- **Nodes can’t detect collisions:** Makes collisions expensive and to be avoided



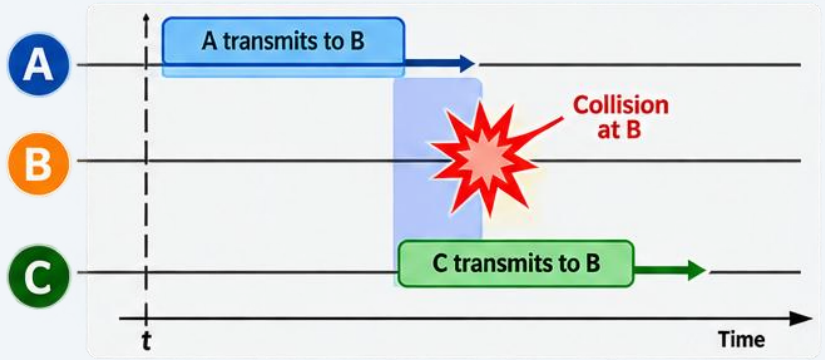
Wireless LAN (IEEE 802.11)

Hidden Node Problem

(Nodes cannot hear each other, but can both reach the receiver)

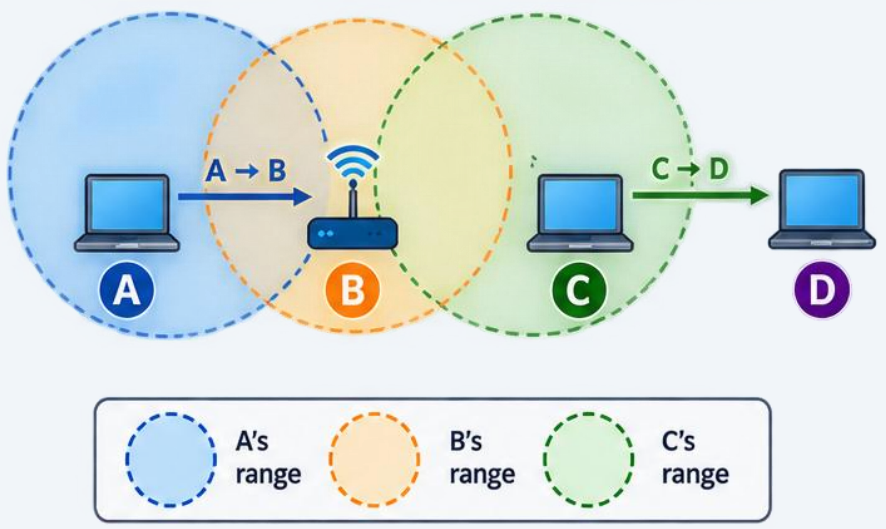


Problem: A and C cannot sense each other, so both may transmit to B at the same time → **collision at B.**

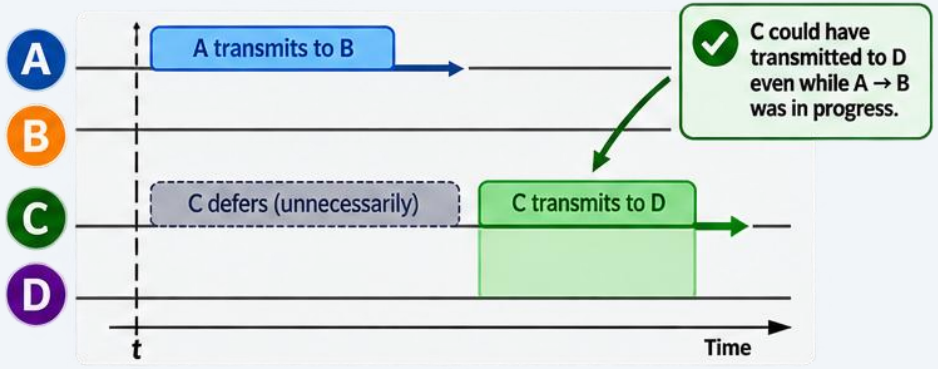


Exposed Node Problem

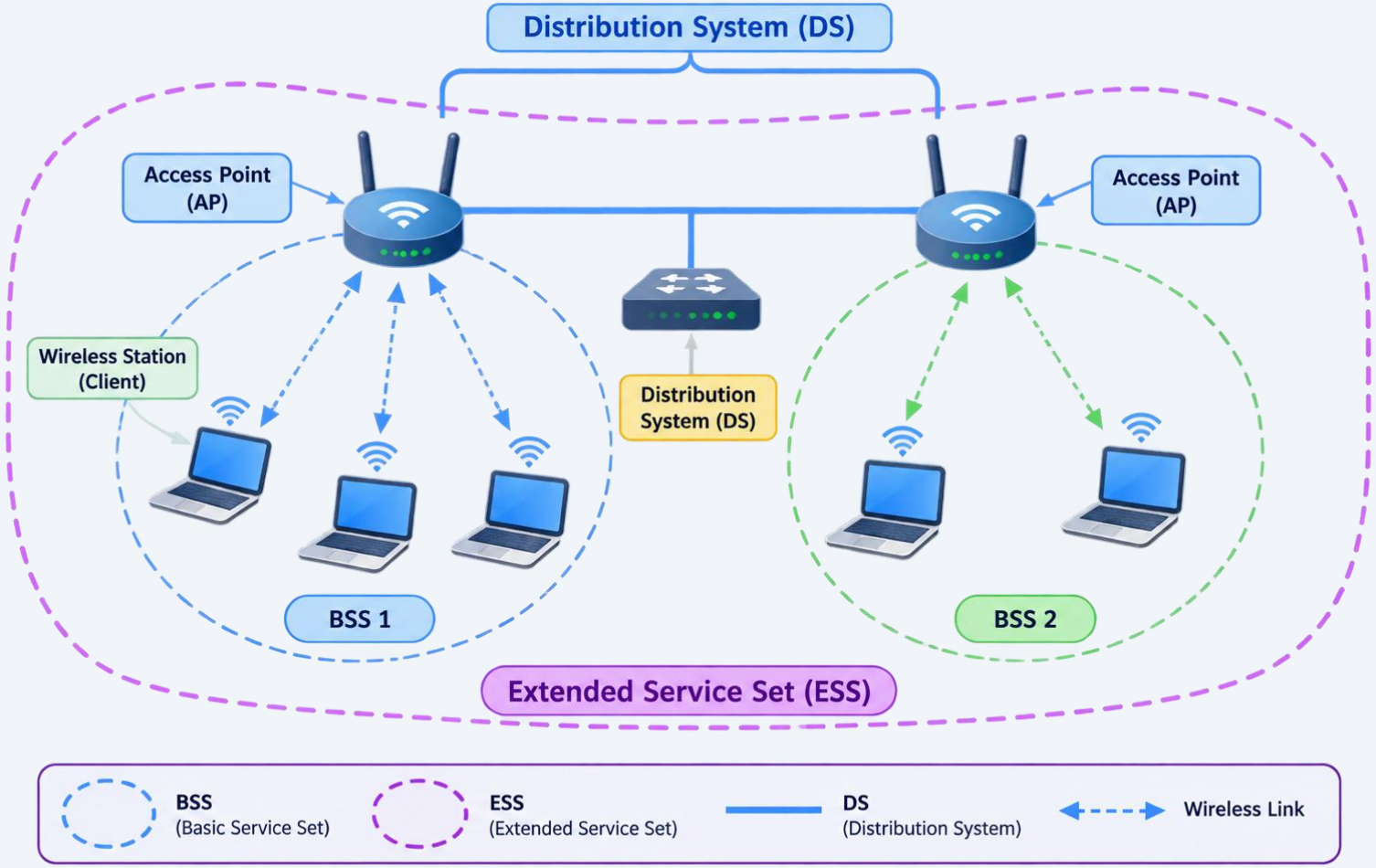
(Node defers unnecessarily, though it can transmit without causing interference)



Problem: C can hear B, so it defers transmission (thinks channel is busy), even though it can transmit to D without causing interference.

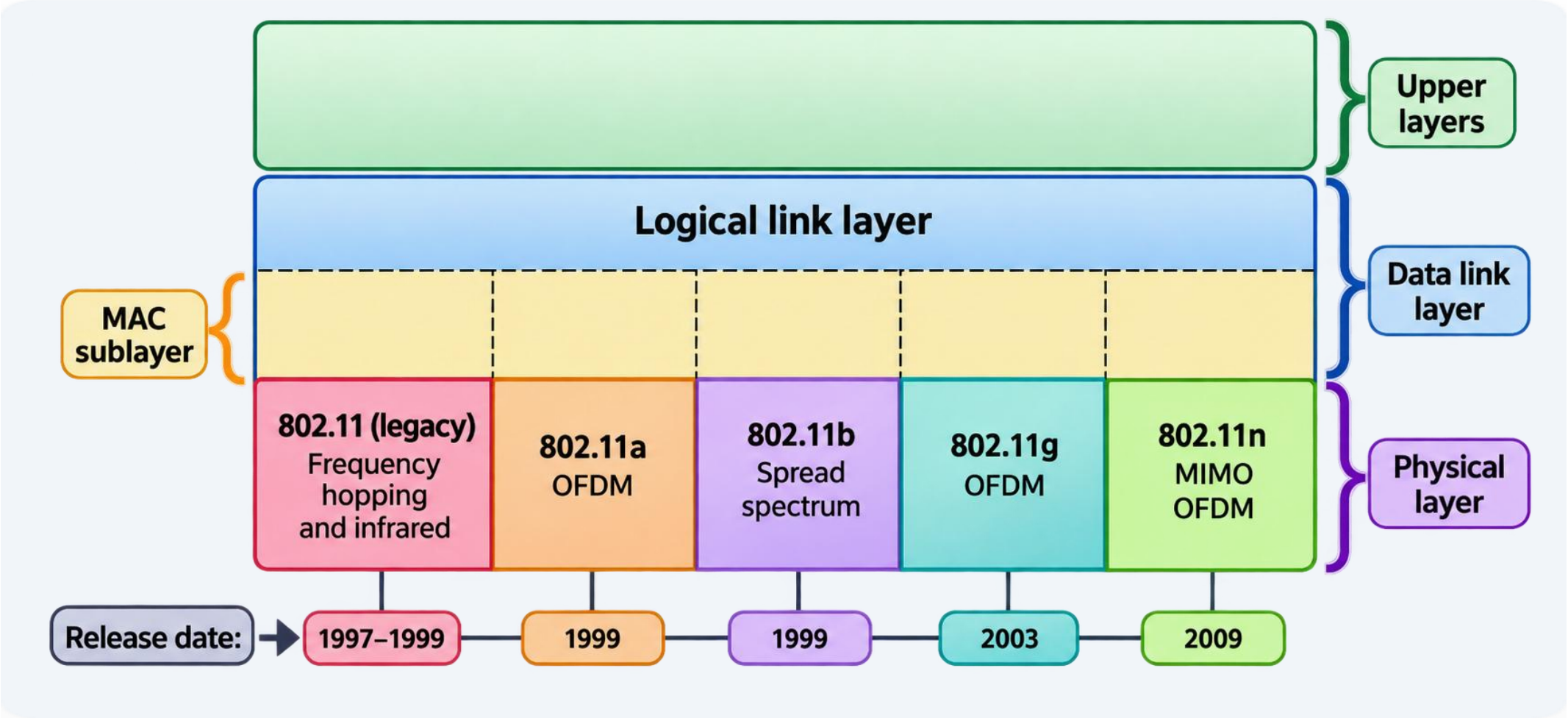


802.11 Architecture/Protocol Stack



802.11 Architecture/Protocol Stack

The same MAC Protocol is used across different physical layers



802.11 Architecture/Protocol Stack

NICs are compatible with multiple physical layers E.g., 802.11 a/b/g

Name	Technique	Max. Bit Rate
802.11b	Spread spectrum, 2.4 GHz	11 Mbps
802.11g	OFDM, 2.4 GHz	54 Mbps
802.11a	OFDM, 5 GHz	54 Mbps
802.11n	OFDM with MIMO, 2.4/5 GHz	600 Mbps

802.11 - MAC layer I - DFWMAC

Traffic Services

- Asynchronous Data Service (mandatory)
  - exchange of data packets based on “best-effort”
  - support of broadcast and multicast
- Time-Bounded Service (optional)
  - implemented using PCF (Point Coordination Function)

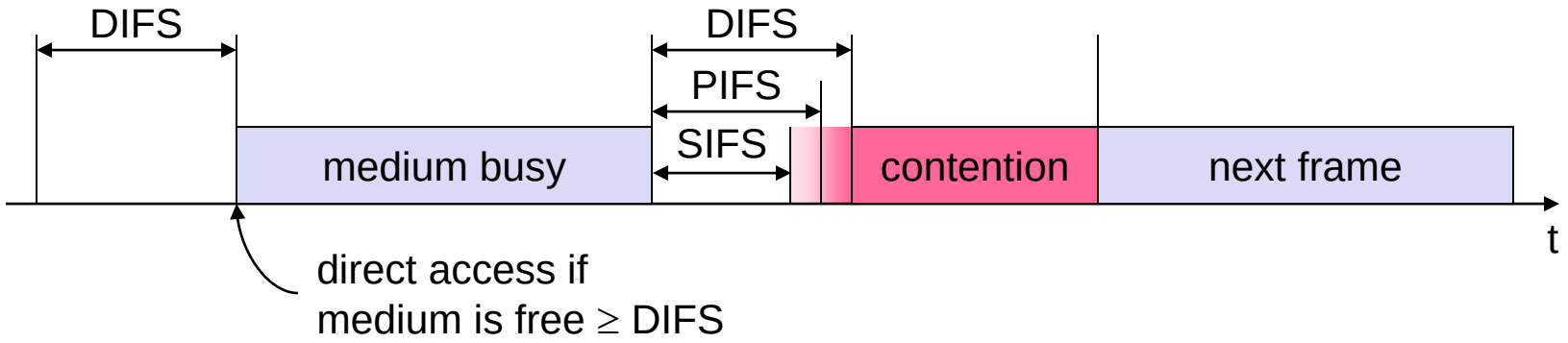
Access methods

- DFWMAC-DCF CSMA/CA (mandatory)
  - collision avoidance via randomized “back-off“ mechanism
  - minimum distance between consecutive packets
  - ACK packet for acknowledgments (not for broadcasts)
- DFWMAC-DCF w/ RTS/CTS (optional)
  - Distributed Foundation Wireless MAC
  - avoids hidden terminal problem
- DFWMAC- PCF (optional)
  - access point polls terminals according to a list

802.11 - MAC layer I - DFWMAC

Priorities

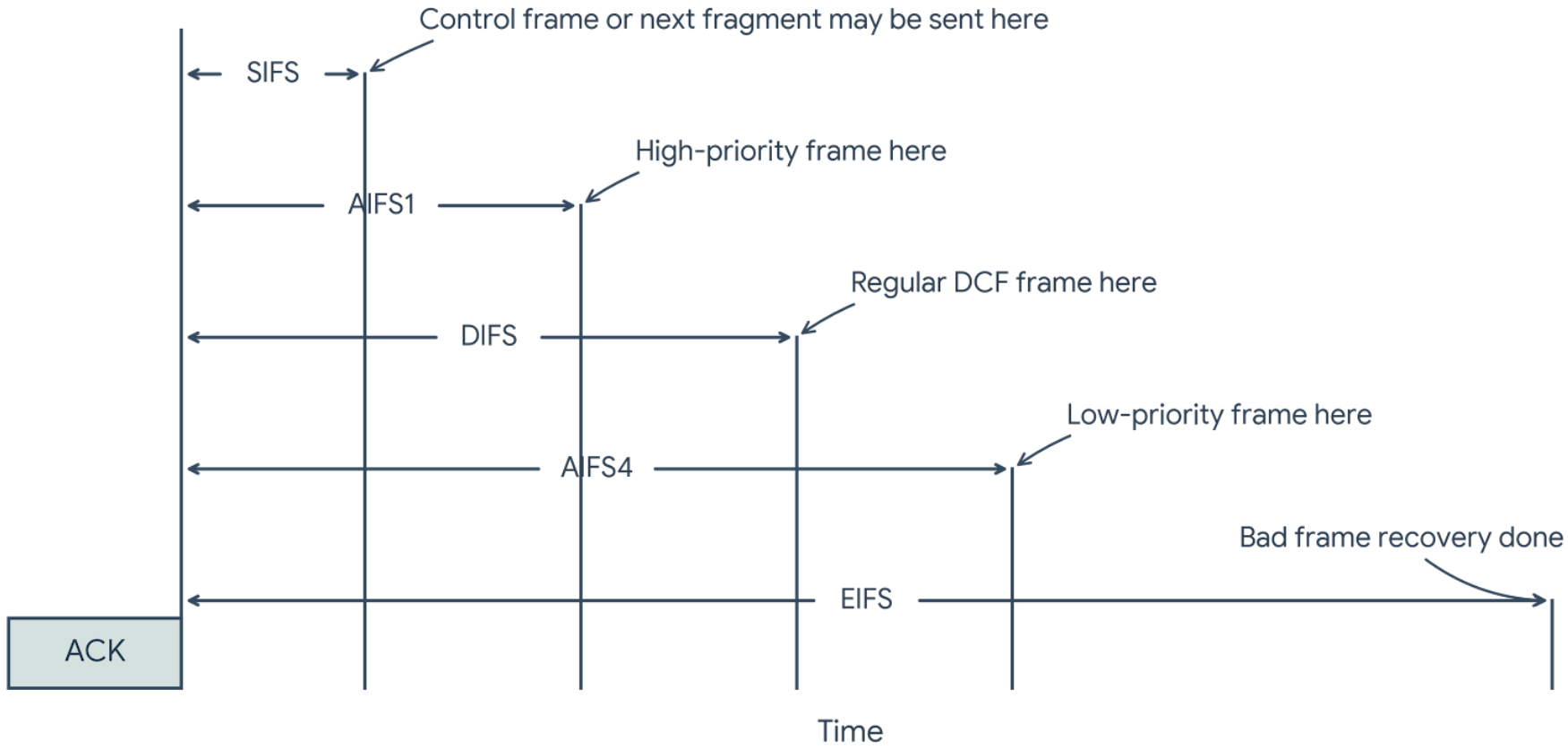
- Defined through different inter-frame spaces
- no guaranteed, hard priorities
- SIFS (Short Interframe Spacing)
  - highest priority, for ACK, CTS, polling response
- PIFS (PCF IFS)
  - Medium priority, for time-bounded service using PCF
- DIFS (DCF, Distributed Coordination Function IFS)
  - Lowest priority, for asynchronous data service



802.11 - MAC layer I - DFWMAC

Priorities

- AIFS1 is same as PIFS while AIFS4 is used for background traffic that can be deferred
- until after regular traffic
- EIFS is used only by a station that has just received a bad or unknown frame, to report the problem.



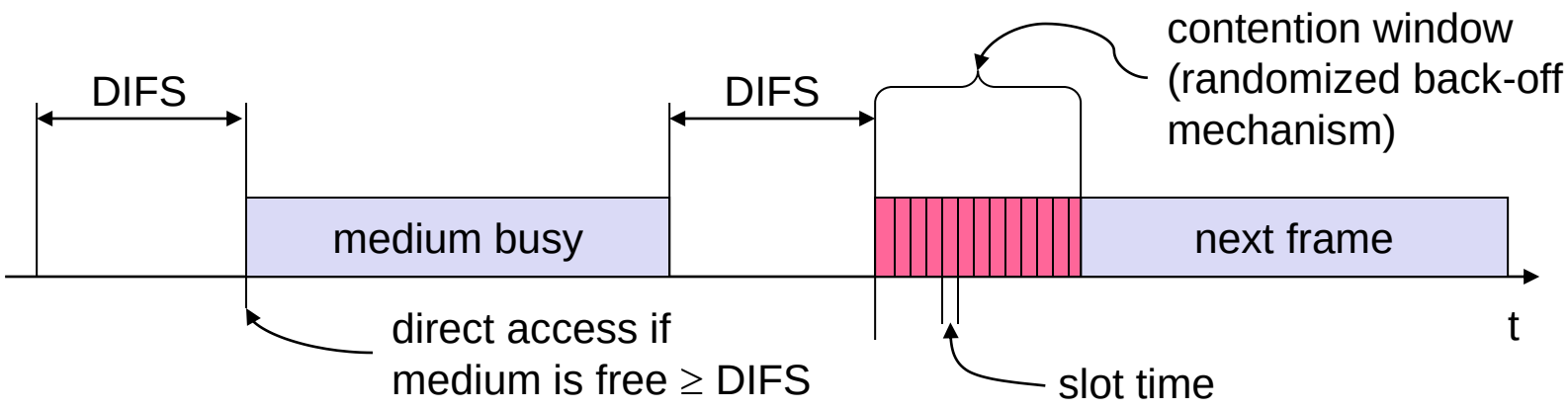
802.11 - MAC layer I - DFWMAC

Short Interframe Space (SIFS) values

WLAN Physical layer	SIFS value
FHSS	28 µs
DSSS	10 µs
OFDM	6 µs
HR/DSSS	10 µs
ERP	10 µs

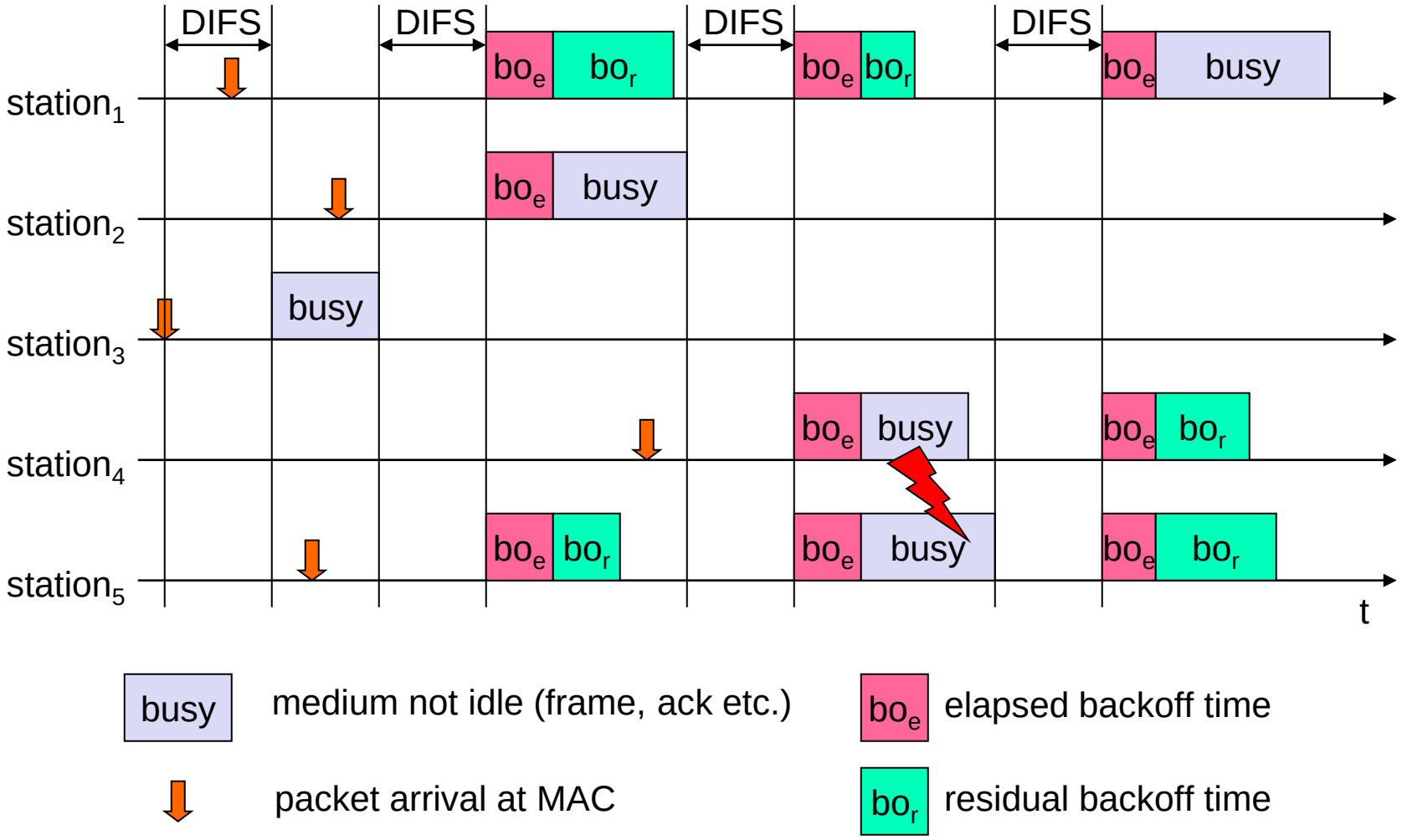
- SIFSTime = aRxRFDelay + aRxPLCPDelay + aMACProcessingDelay + aRxTxTurnaroundTime
- PIFS = SIFSTime + SlotTime

802.11 - CSMA/CA access method I



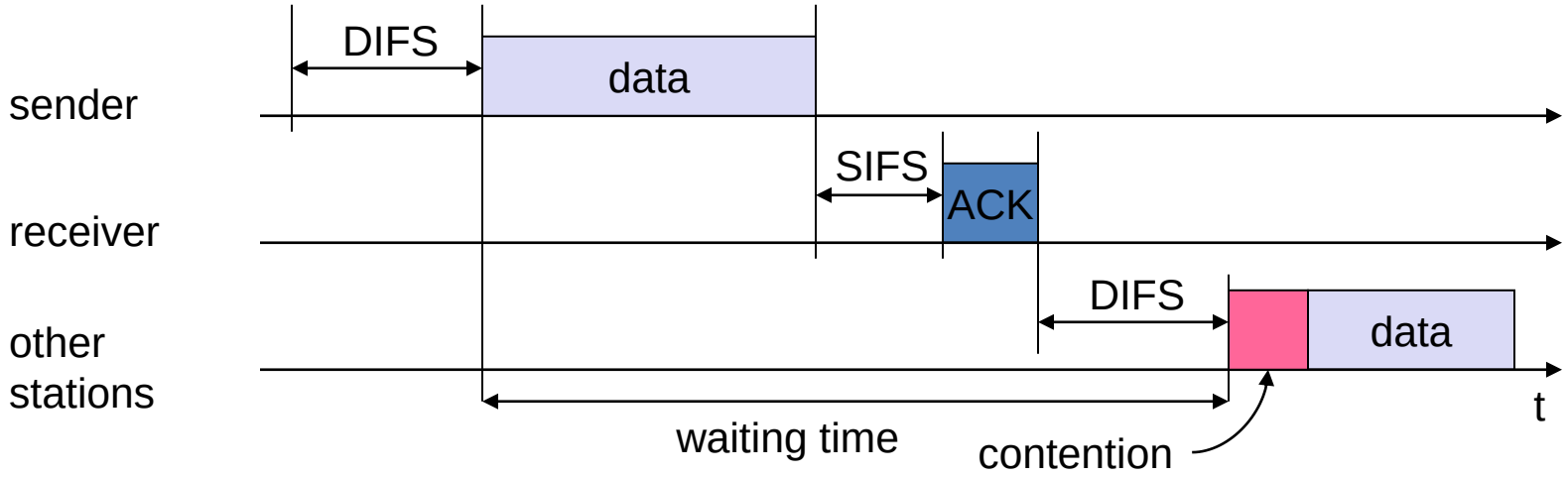
- Station ready to send starts sensing the medium (Carrier Sense based on CCA, Clear Channel Assessment)
- If the medium is free for the duration of an Inter-Frame Space (IFS), the station can start sending (IFS depends on service type)
- If the medium is busy, the station has to wait for a free IFS, then the station must additionally wait a random back-off time (collision avoidance, multiple of slot-time)
- If another station occupies the medium during the back-off time of the station, the back-off timer stops (fairness)

802.11 - competing stations - simple version



802.11 - CSMA/CA access method I

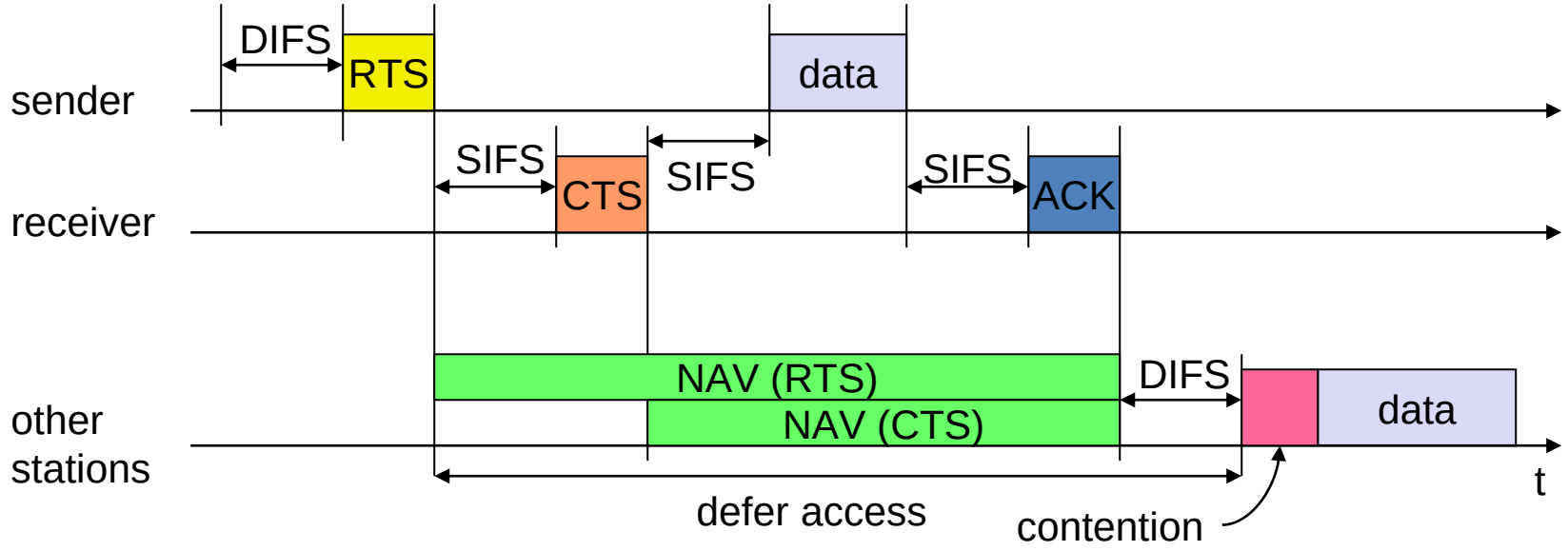
Sending unicast packets



- Station has to wait for DIFS before sending data
- Receivers acknowledge at once (after waiting for SIFS) if the packet was received correctly (CRC)
- Automatic retransmission of data packets in case of transmission errors

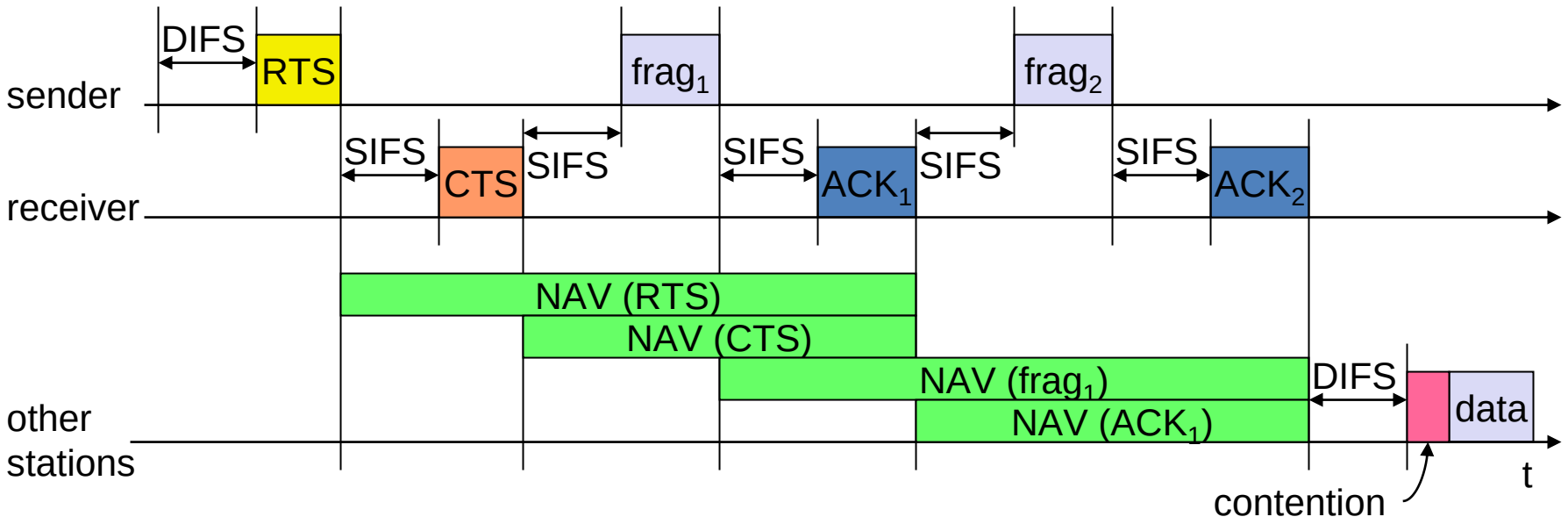
802.11 - CSMA/CA access method II

Sending unicast packets

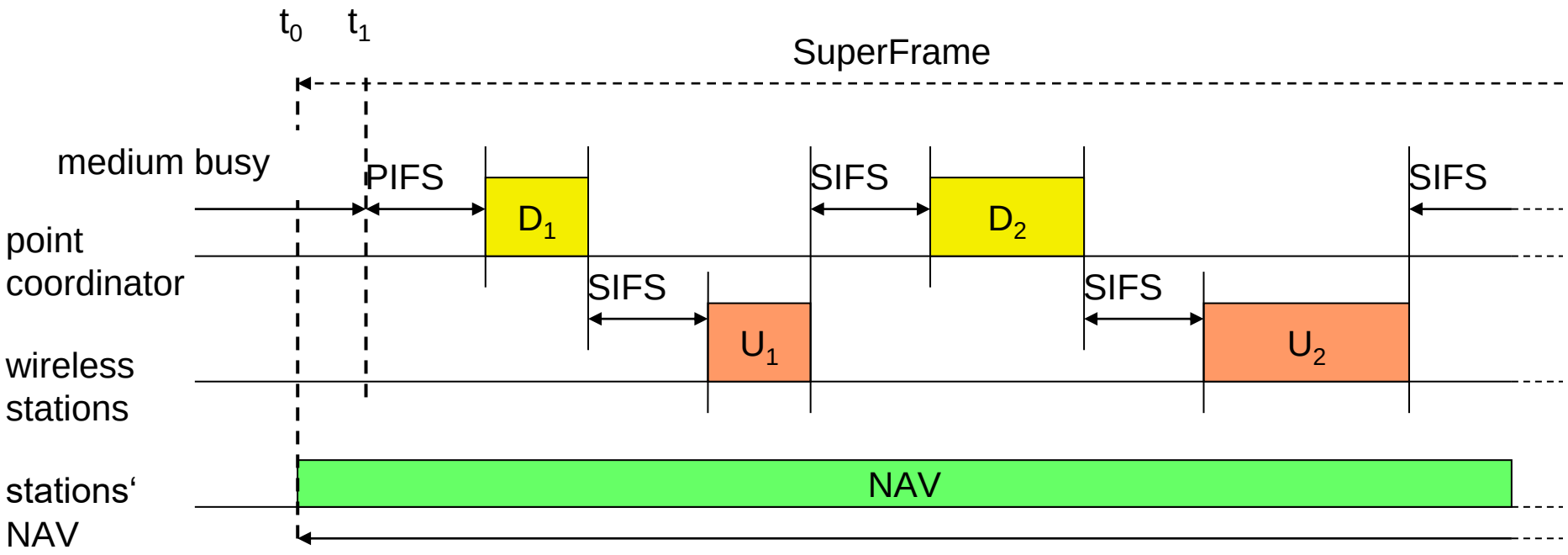


- station can send RTS with reservation parameter after waiting for DIFS (reservation determines amount of time the data packet needs the medium)
- acknowledgement via CTS after SIFS by receiver (if ready to receive)
- sender can now send data at once, acknowledgement via ACK
- other stations store medium reservations distributed via RTS and CTS

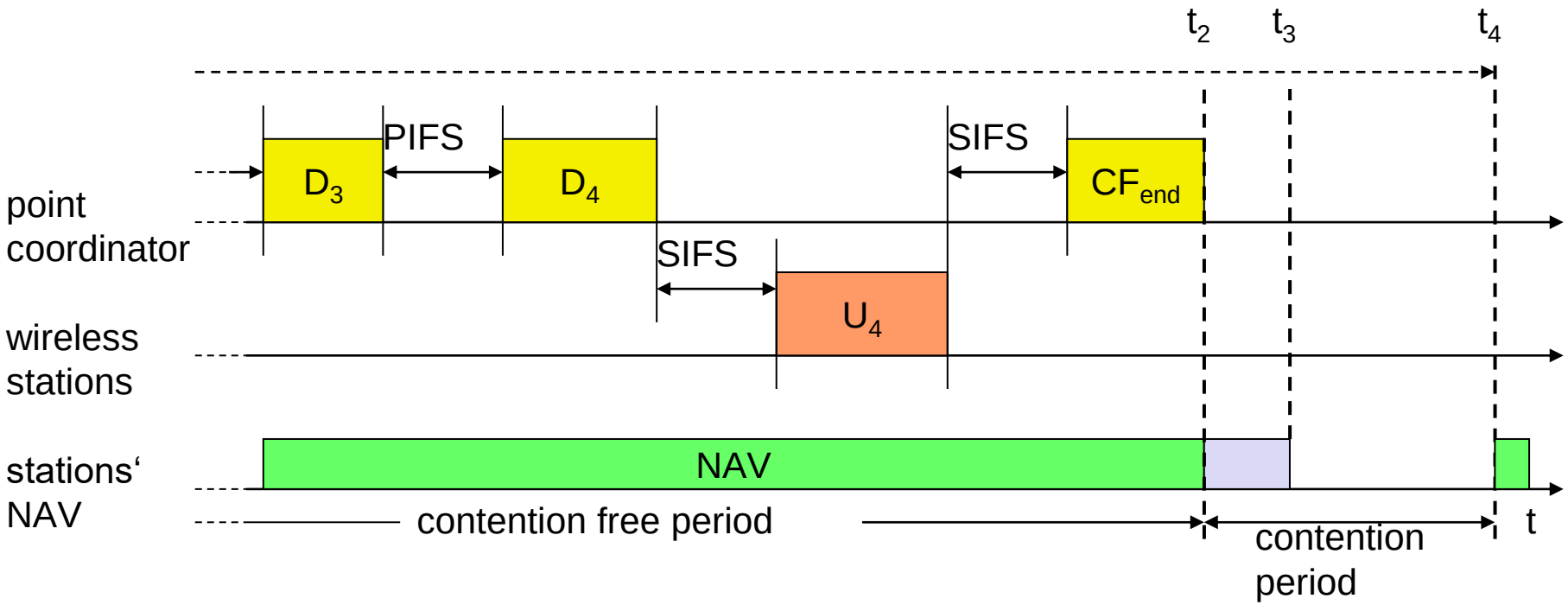
802.11 - CSMA/CA - Fragmentation



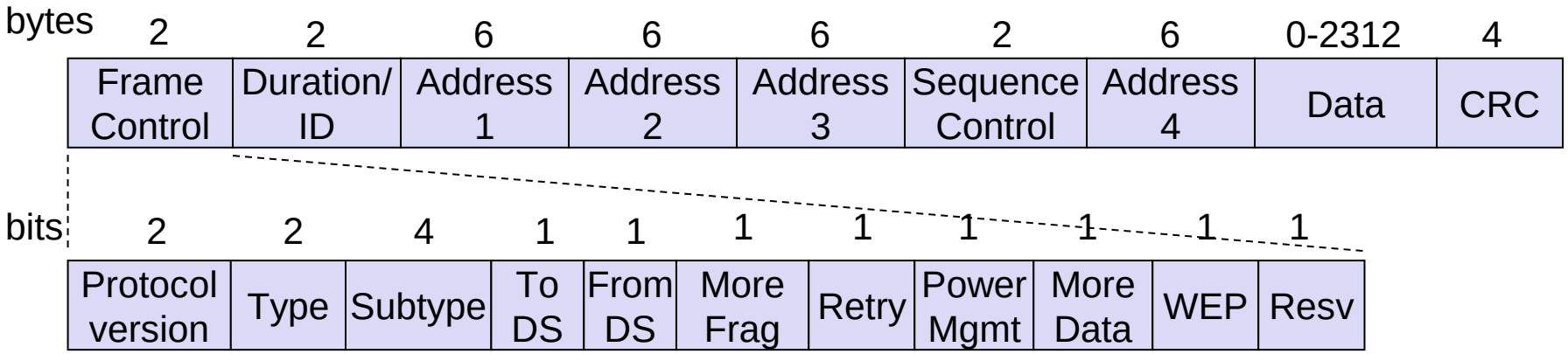
802.11 - CSMA/CA – PCF 1



802.11 - CSMA/CA – PCF 2



802.11 – Frame Format



- Types
  - Control frames, management frames, data frames
- Sequence numbers
  - Important against duplicated frames due to lost ACKs
- Addresses
  - receiver, transmitter (physical), BSS identifier, sender (logical)
- Miscellaneous
  - Sending time, checksum, frame control, data

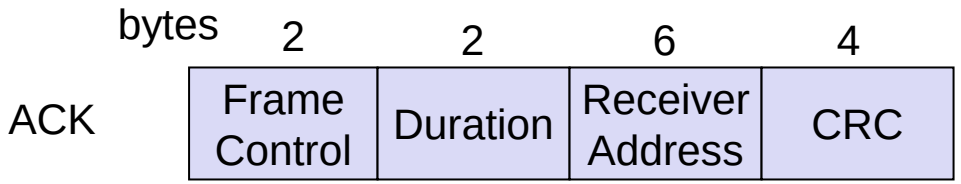
802.11 – Address Format

scenario	to DS	from DS	address 1	address 2	address 3	address 4
ad-hoc network	0	0	DA	SA	BSSID	-
infrastructure network, from AP	0	1	DA	BSSID	SA	-
infrastructure network, to AP	1	0	BSSID	SA	DA	-
infrastructure network, within DS	1	1	RA	TA	DA	SA

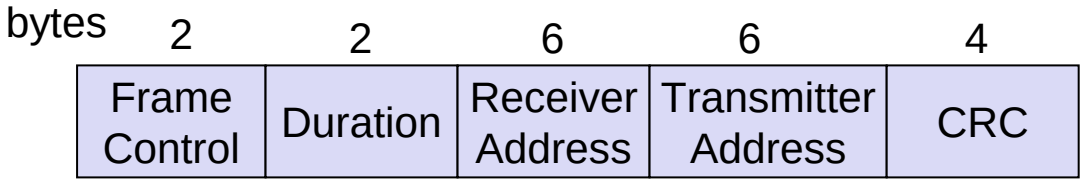
DS: Distribution System
AP: Access Point
DA: Destination Address
SA: Source Address
BSSID: Basic Service Set Identifier
RA: Receiver Address
TA: Transmitter Address

Special Frames: ACK, RTS, CTS

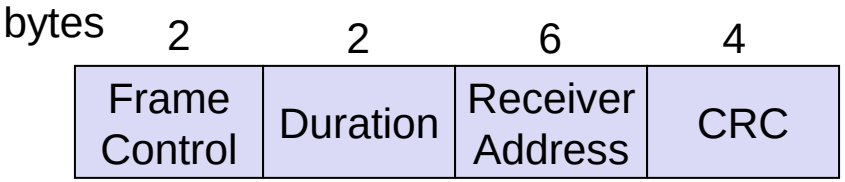
Acknowledgement



Request To Send



Clear To Send



802.11 - MAC management

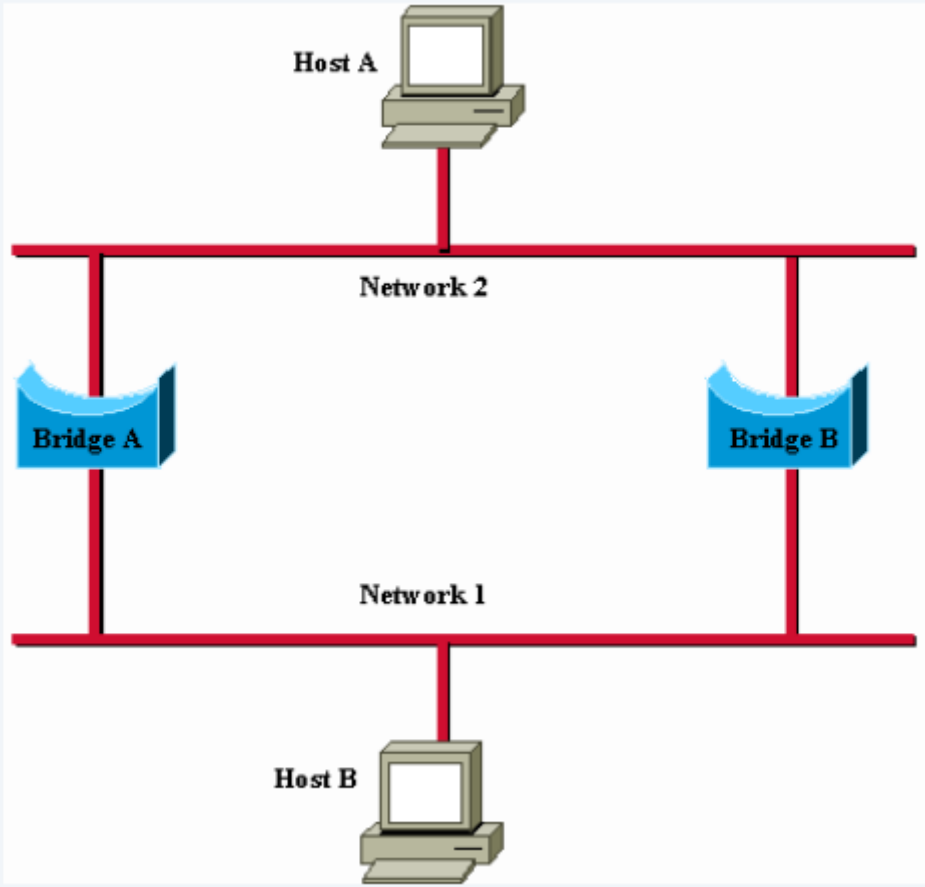
- Synchronization
  - Try to find a LAN, try to stay within a LAN
  - Timer, etc.
- Power management
  - Sleep mode without missing a message
  - Periodic sleep, frame buffering, traffic measurements
- Association/Reassociation
  - Integration into a LAN
  - Roaming, i.e. change networks by changing access points
  - Scanning, i.e. active search for a network
- MIB - Management Information Base
  - Managing, read, write

Data Link Layer Switching

Transparent Bridges & Learning

Bridges and switches connect multiple LAN segments, operating entirely at the Data Link layer by inspecting MAC addresses.

- **Backward Learning Algorithm:** Bridges observe incoming frames to associate source MAC addresses with specific input ports.
- **Intelligent Forwarding:** Frames are only sent out the port associated with the destination address, saving bandwidth on other segments.
- **Flooding:** If the destination port is unknown, the frame is broadcast to all ports (except the origin).



Data Link Layer Switching

Learning Bridge

Gradual building of Table

Address	Port

a. Original

Address	Port
71:2B:13:45:61:41	1

b. After A sends a frame to D

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4

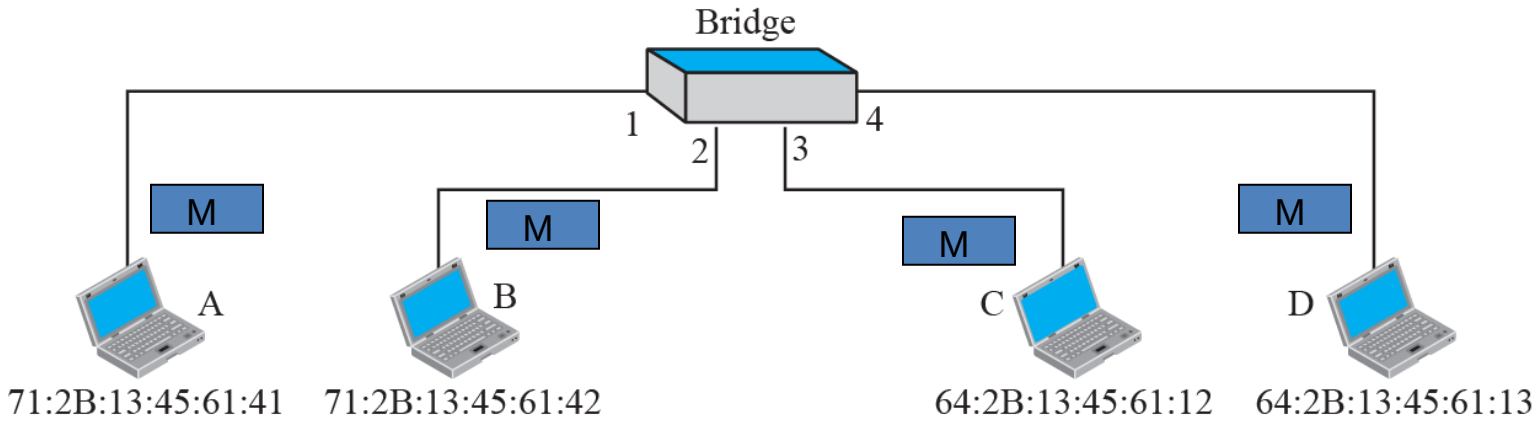
c. After D sends a frame to B

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2

d. After B sends a frame to A

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3

e. After C sends a frame to D



Data Link Layer Switching

Learning Bridge

The backward learning algorithm picks the output port:

- Associates the source address on the frame with input port
- Frame with destination address is sent to learned port
- Unlearned destinations are sent to all other ports

Needs no configuration

- Forgets unused addresses to allow changes
- Bandwidth efficient for two-way traffic

Data Link Layer Switching

Forwarding Procedure

1. If the port for the destination address is the same as the source port, discard the frame.
2. If the port for the destination address and the source port are different, forward the frame to the destination port.
3. If the destination port is unknown, use flooding and send the frame on all ports except the source port.

This algorithm needs to be applied for each frame, hence, it is implemented in special-purpose VLSI chips

Data Link Layer Switching

Bridge Forwarding Types

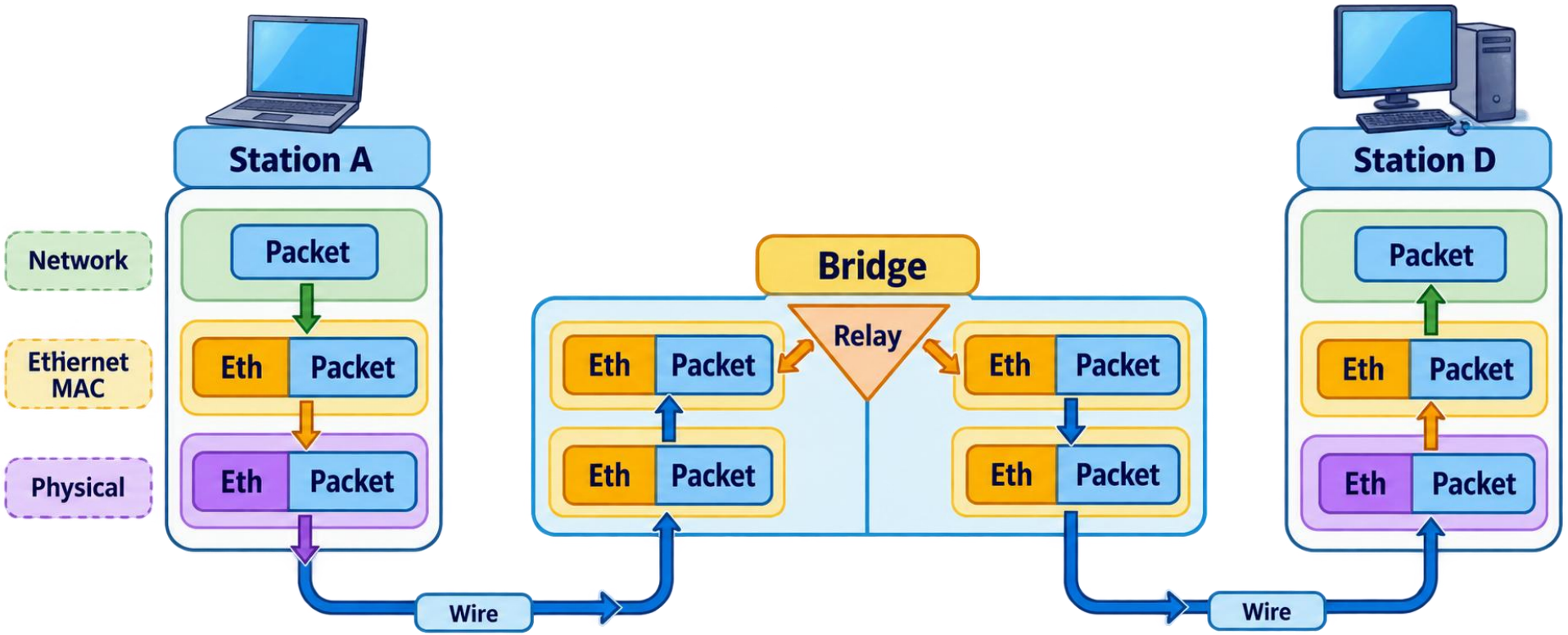
- A store-and-forward switch stores each incoming frame in its entirety, then examines it and forwards it.
- A cut-through switch starts to forward incoming frames before they have arrived completely. As soon as the destination address is in, forwarding can begin.
- Store-and-forward switches store entire frames before forwarding them. After a frame comes in, the checksum can be verified. If the frame is damaged, the switch discards it immediately.
- With cut-through, damaged frames cannot be discarded by the switch because, by the time the error is detected, the frame is already gone.

Data Link Layer Switching

Learning Bridge

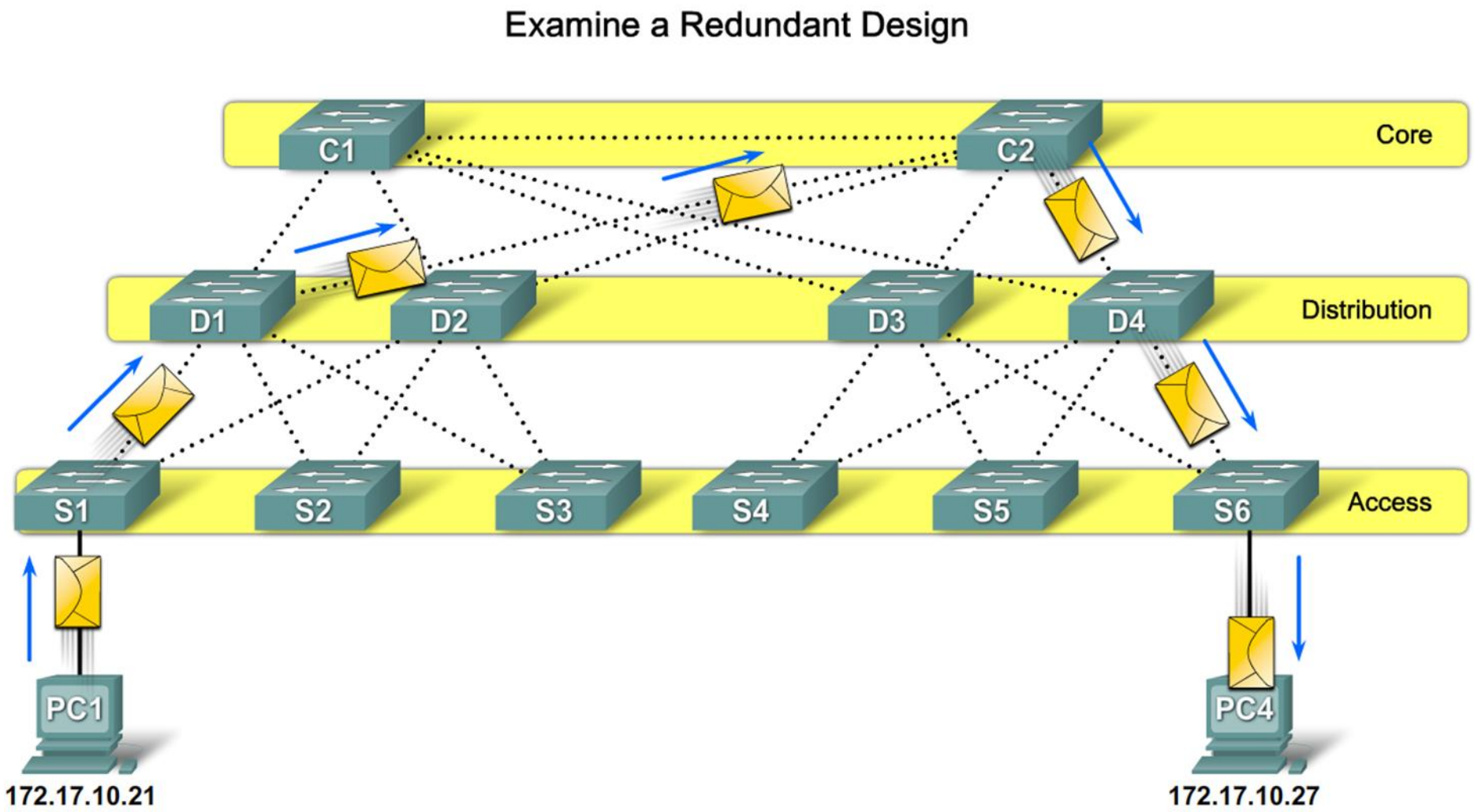
Bridges extend the Link layer:

- Use but don’t remove the Ethernet header/addresses
- Do not inspect the Network header



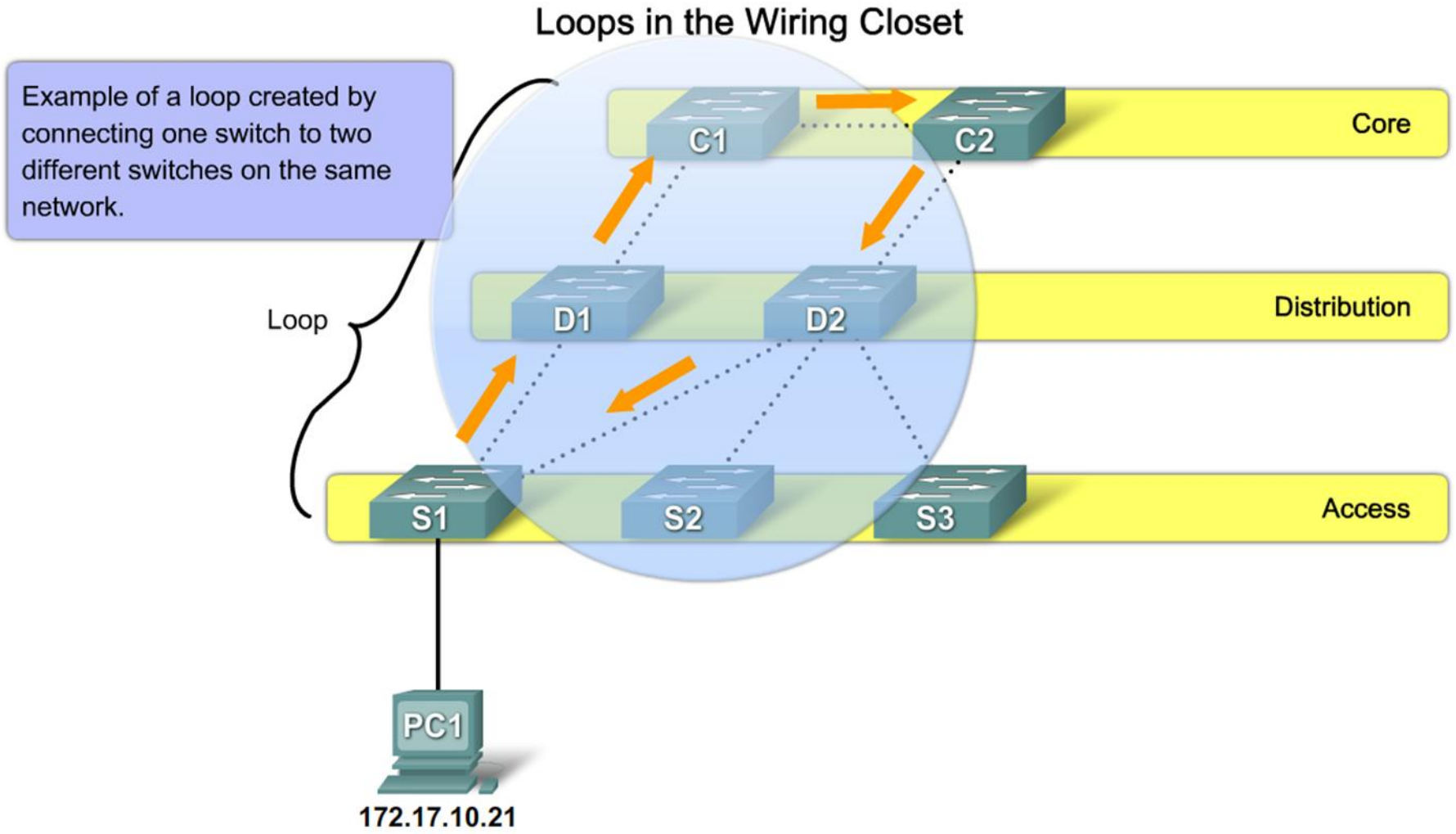
Data Link Layer Switching

Spanning Tree – The need for redundancy (The problem)



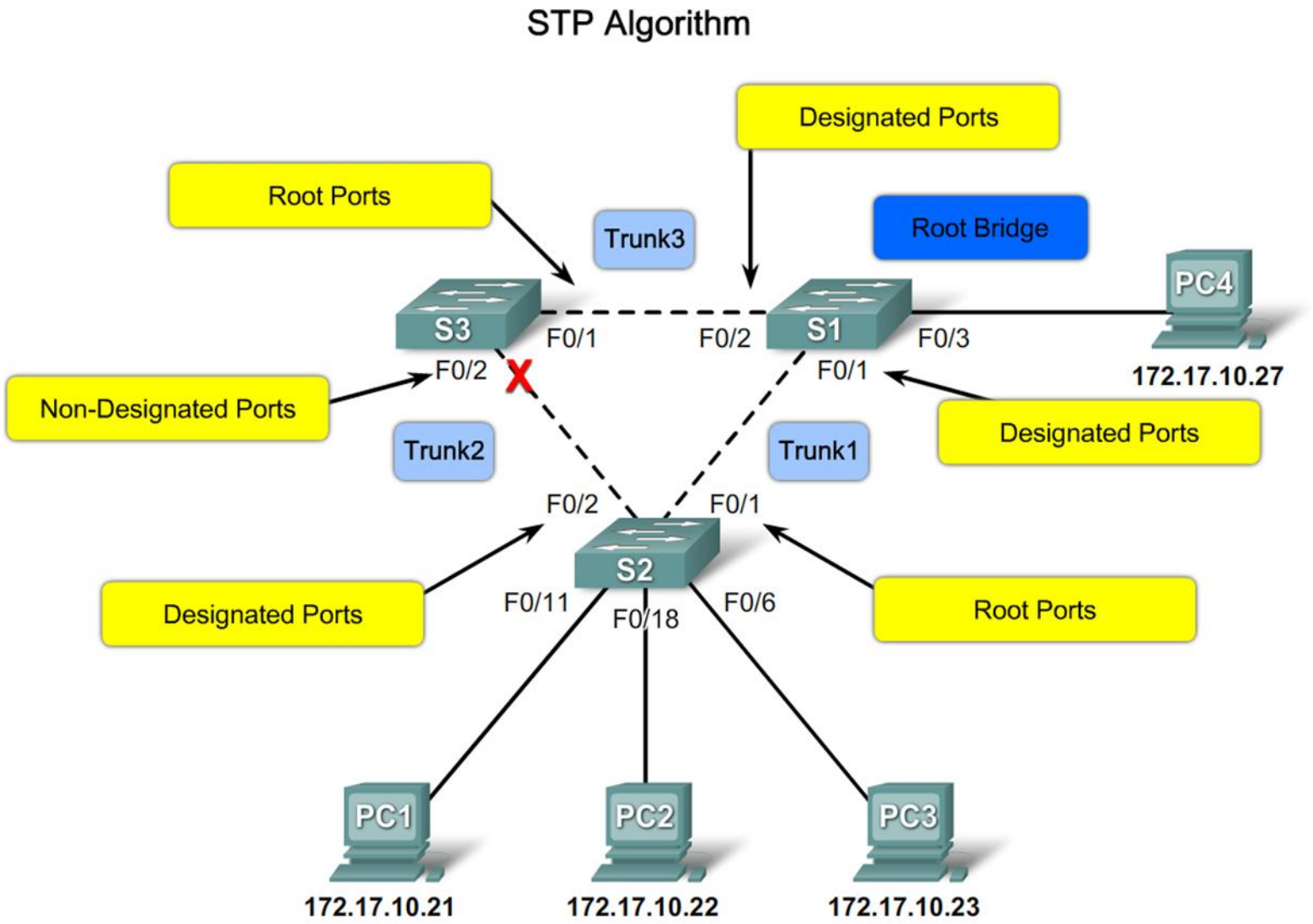
Data Link Layer Switching

Spanning Tree – The need for redundancy (The problem)



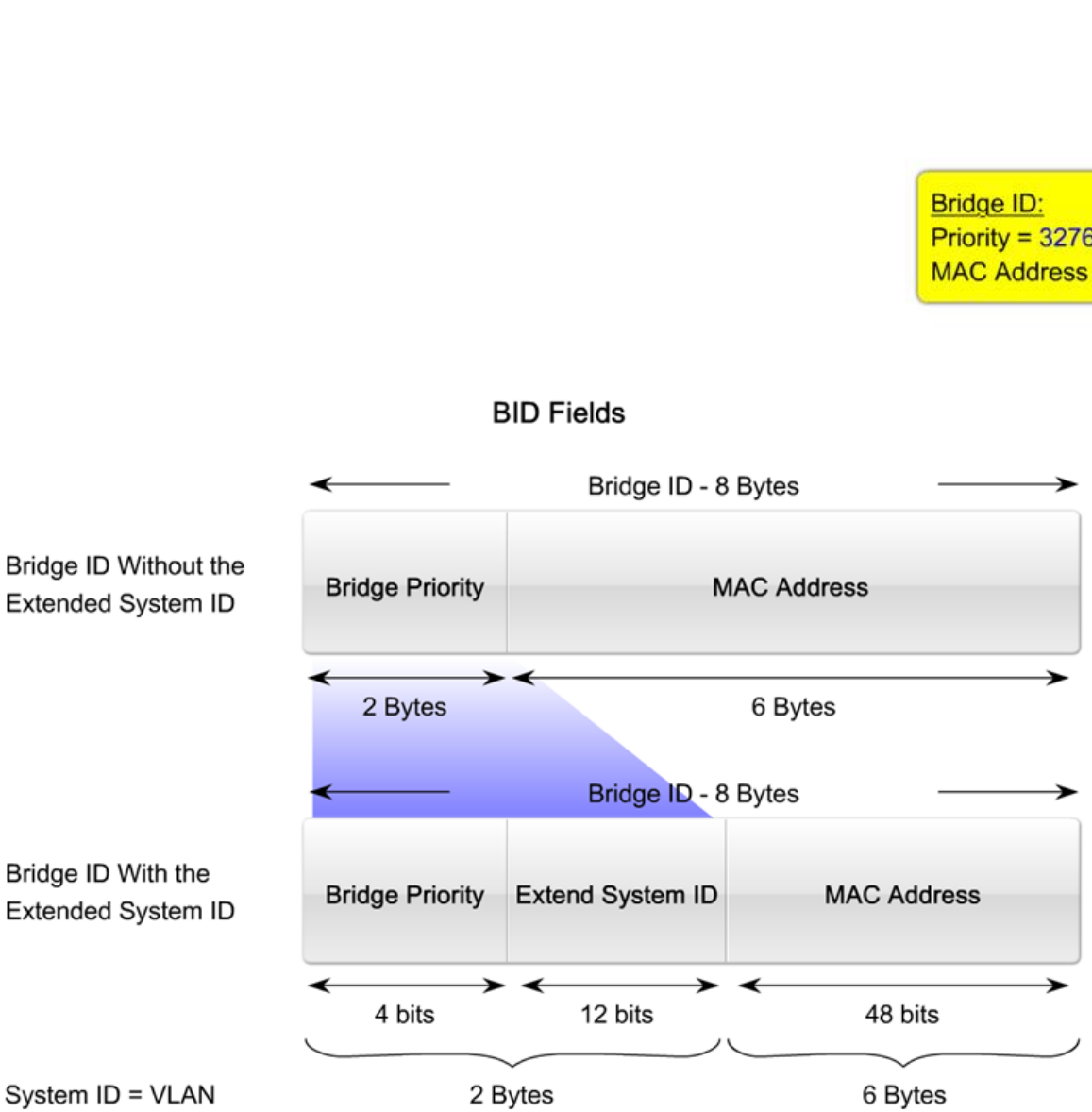
Data Link Layer Switching

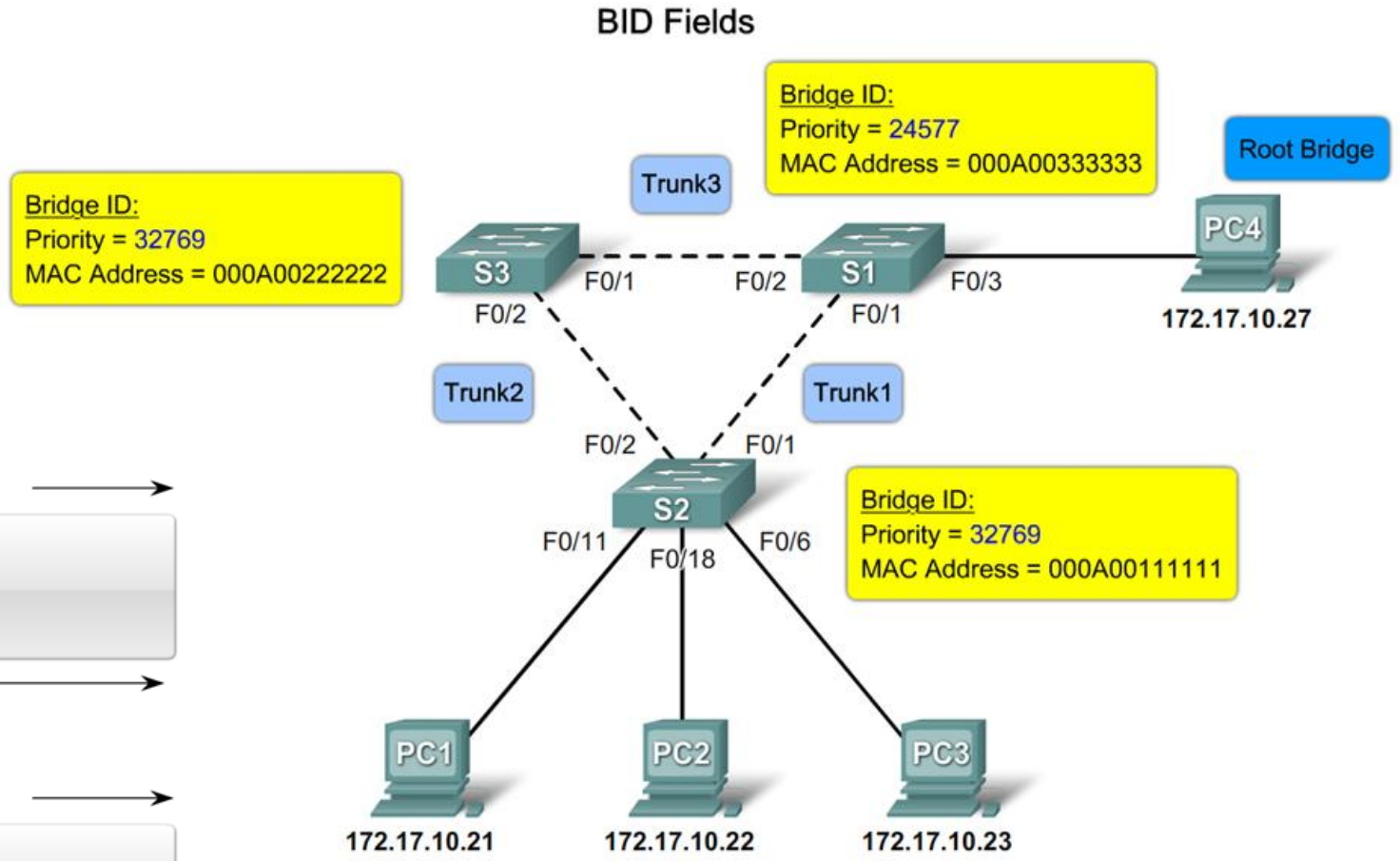
Spanning Tree – Eliminate Layer 2 Loops in Network



Data Link Layer Switching

Spanning Tree – Eliminate Layer 2 Loops in Network

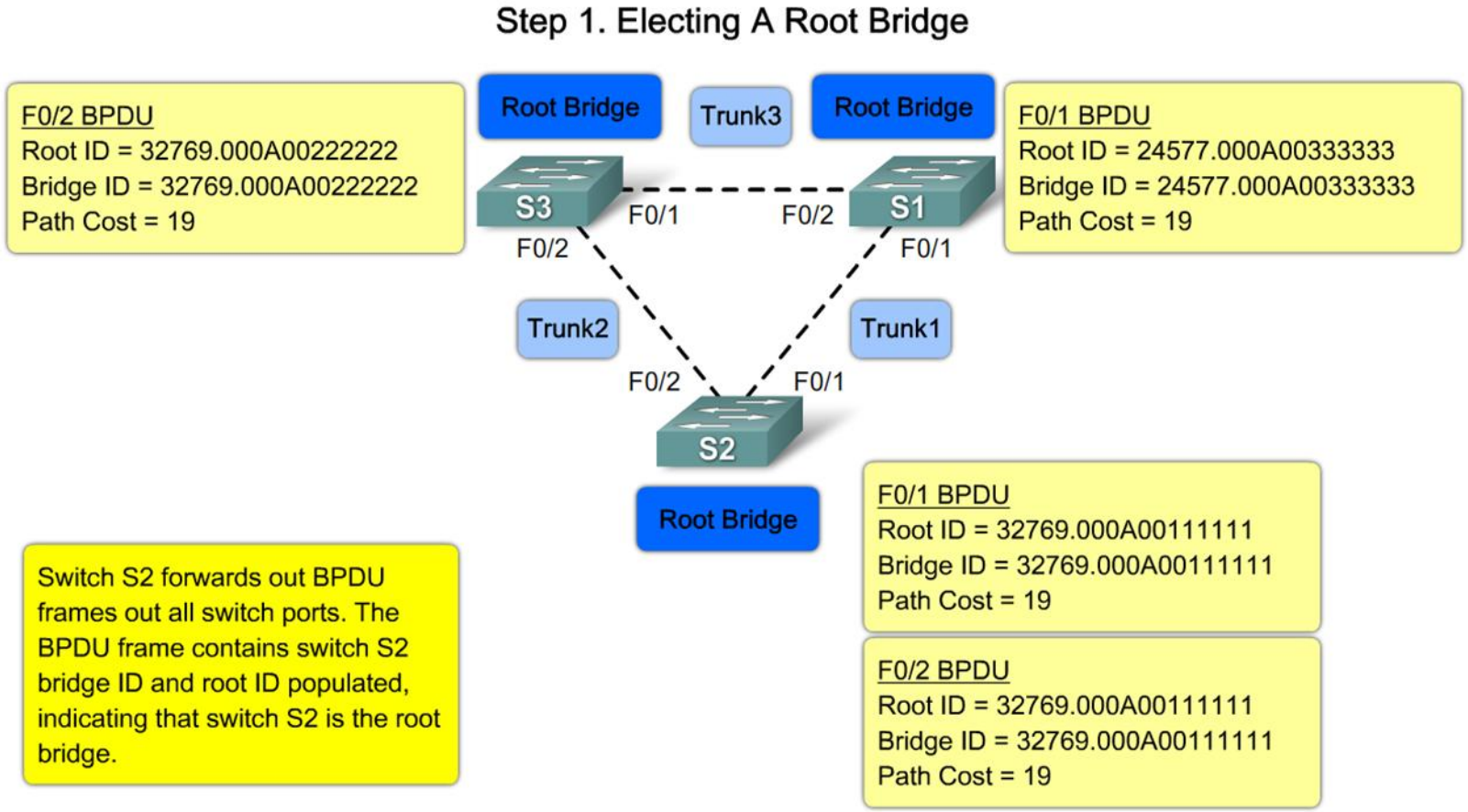




Data Link Layer Switching

STP Algorithm – Step 1

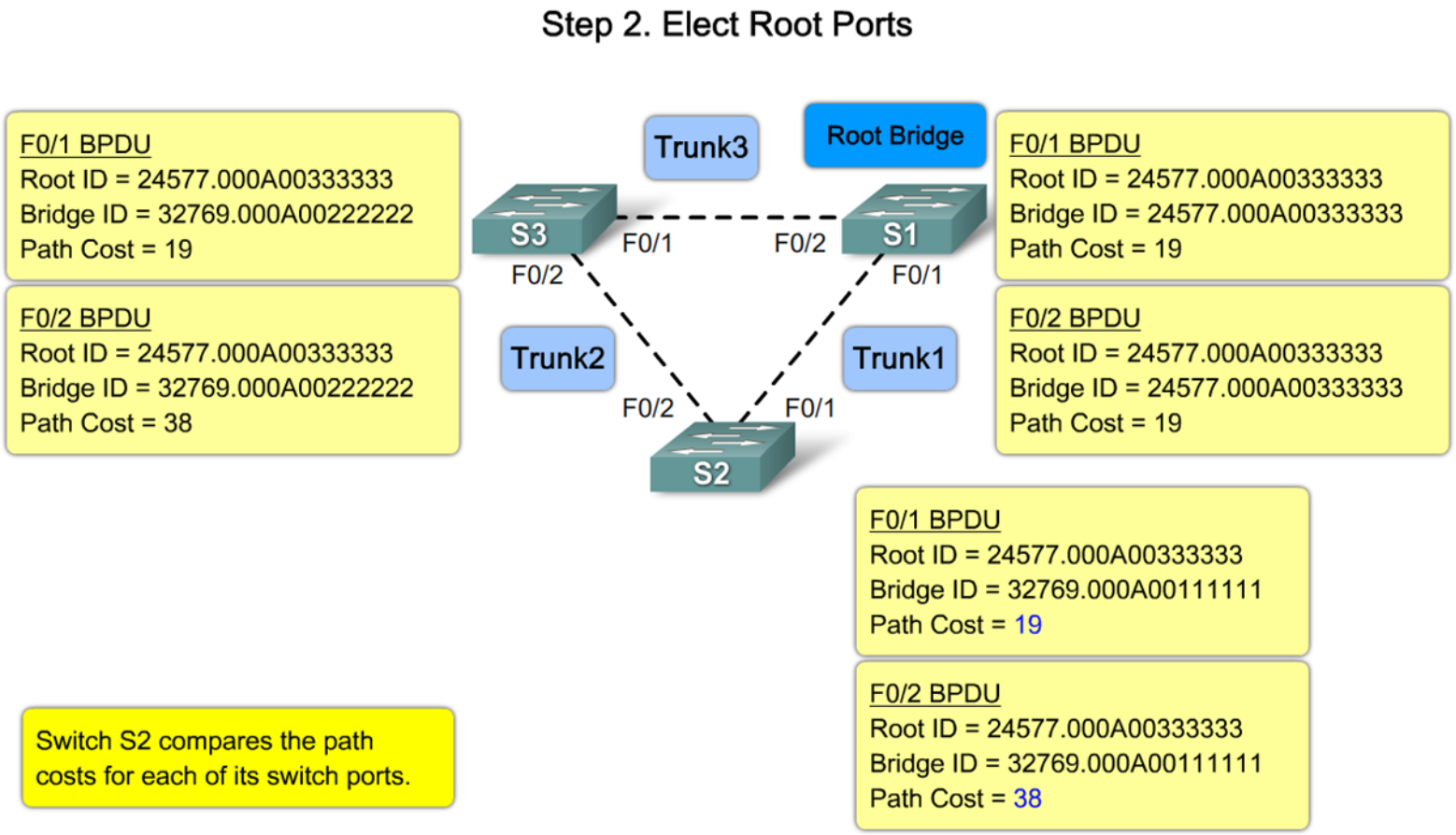
- The bridge with the lowest Bridge ID (BID) becomes the root bridge.
- BPDUs containing the BID are exchanged between the switches to elect a root bridge
- All ports on the root bridge are set as designated and thus are always set to a forwarding state.



Data Link Layer Switching

STP Algorithm – Step 2

- Elect Root Port: For non-root bridges, there will be only one root port. The root port will be the port with the lowest path cost to the root bridge. The root port will also be set to the forwarding state
- Cost of the path can be changed with spanning-tree cost N command

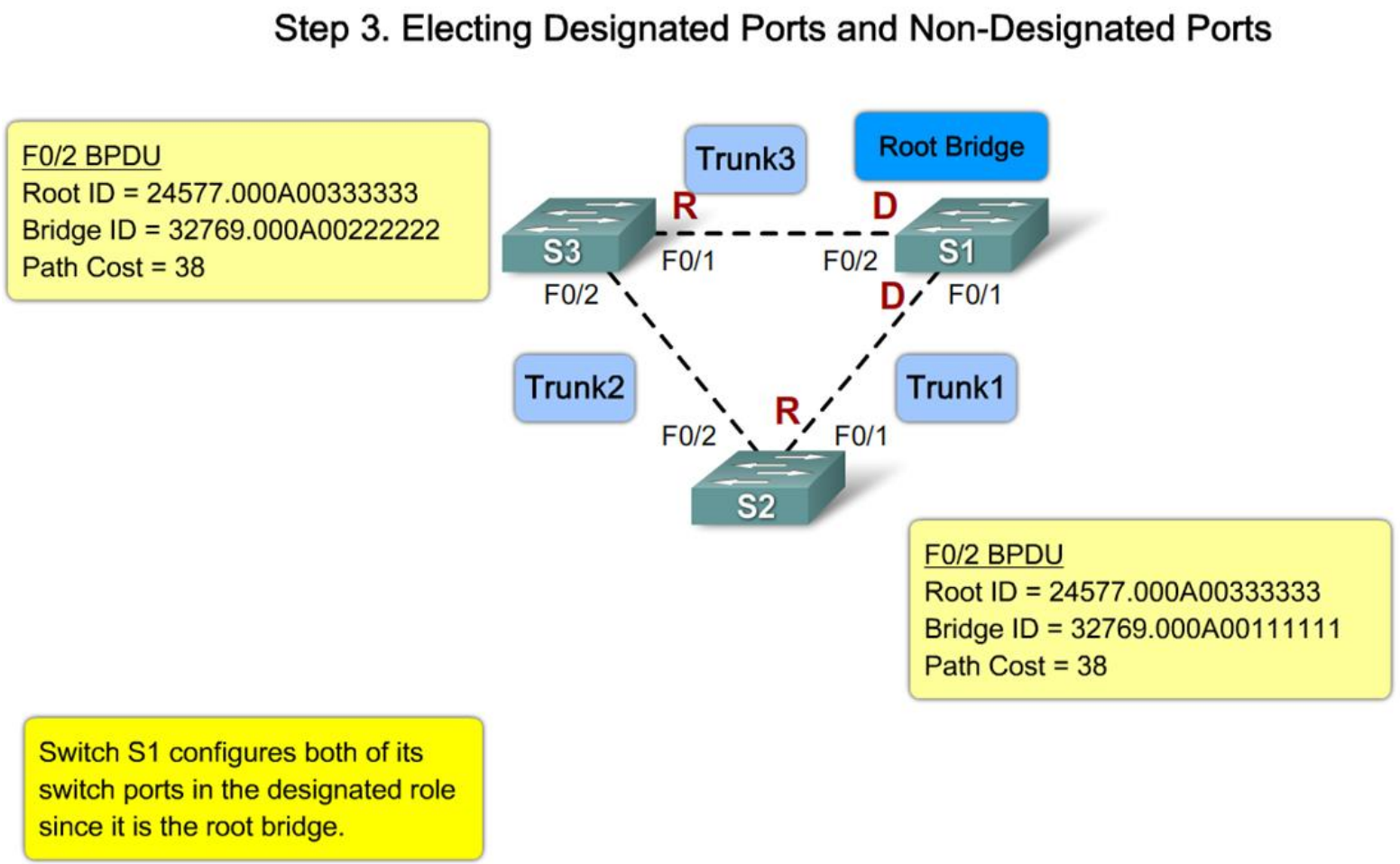


Link Speed	STP cost
4 Mbps	250
10 Mbps	100
16 Mbps	62
100 Mbps	19
1 Gbps	4
10 Gbps	2

Data Link Layer Switching

STP Algorithm – Step 3

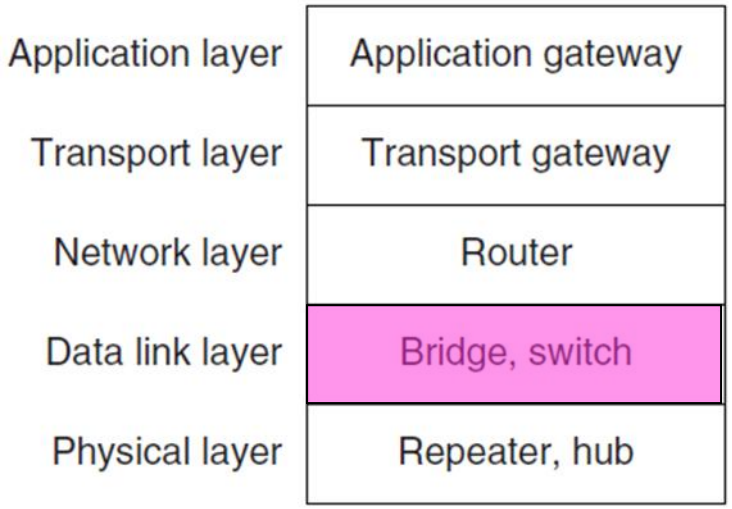
- Electing designated ports and non-designated ports on a switch



Data Link Layer Switching

Repeaters, Hubs, Bridges, Switches, Routers, & Gateways

- Devices are named according to the top most layer they process
- A bridge or LAN switch operates in the Link layer



Virtual LANs (VLANs) - IEEE 802.1Q

Logical Segmentation

VLANs split a single physical LAN into multiple logical LANs. This improves security, eases administration, and reduces broadcast traffic overhead.

Instead of relying on physical wiring to group departments (e.g., HR, Engineering), network administrators assign specific switch ports to specific "colored" VLANs virtually.

The 802.1Q Tag

To identify which VLAN a frame belongs to as it travels between switches, IEEE 802.1Q inserts a special 4-byte tag into the Ethernet header.

- **Access Ports:** Connect to end devices; switch handles tagging/untagging transparently.
- **Trunk Ports:** Connect switches together; frames retain their 802.1Q tags to preserve VLAN identity across the network backbone.

Virtual LANs (VLANs) - IEEE 802.1Q

Logical Segmentation

VLANs split a single physical LAN into multiple logical LANs. This improves security, eases administration, and reduces broadcast traffic overhead.

Instead of relying on physical wiring to group departments (e.g., HR, Engineering), network administrators assign specific switch ports to specific "colored" VLANs virtually.

The 802.1Q Tag

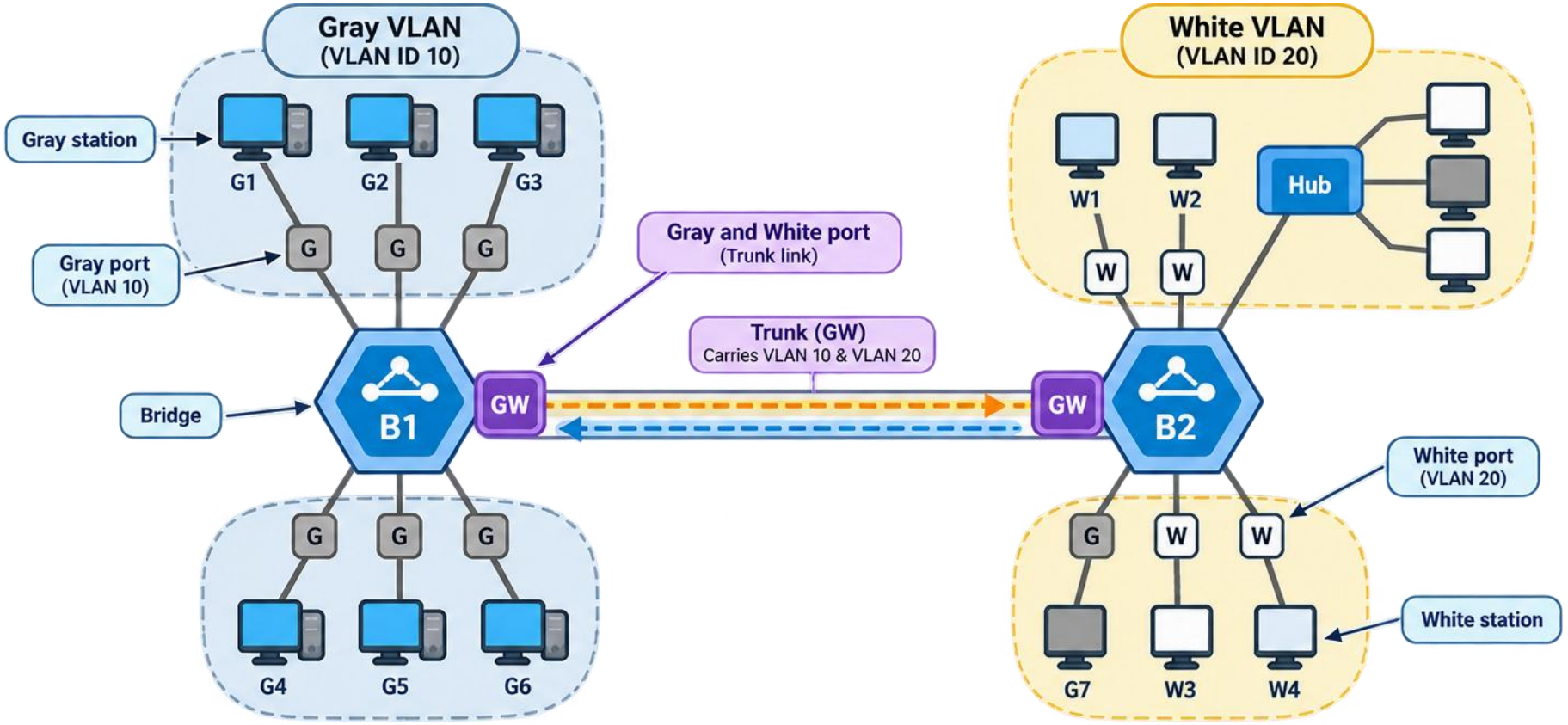
To identify which VLAN a frame belongs to as it travels between switches, IEEE 802.1Q inserts a special 4-byte tag into the Ethernet header.

- **Access Ports:** Connect to end devices; switch handles tagging/untagging transparently.
- **Trunk Ports:** Connect switches together; frames retain their 802.1Q tags to preserve VLAN identity across the network backbone.

VLAN Port Modes

- There are two ways in which a machine can be connected to a switch carrying 802.1Q VLAN traffic:
  - via an access port, where VLAN support is handled by the switch (so the machine sees ordinary, untagged Ethernet frames); or
  - via a trunk port, where VLAN support is handled by the attached machine (which sees 802.1Q-tagged Ethernet frames).
- It is also possible to operate a switch port in a hybrid mode, where it acts as an access port for one VLAN and a trunk port for others (so the attached Ethernet segment carries a mixture of tagged and untagged frames)

Virtual LANs (VLANs) - IEEE 802.1Q



1 Packet from Gray VLAN

G1

Frame from G1 (VLAN 10) reaches B1 on a Gray port.

2 Tagged on Trunk

GW

B1 tags the frame with VLAN 10 and sends it on the trunk (GW).

3 Forward to B2

GW

B2 receives the tagged frame and forwards it to Gray ports (VLAN 10).

4 Delivered to Destination

G7

Frame is delivered only to Gray VLAN members (e.g., G7). White VLAN is isolated.

Legend

- G — Gray port (VLAN 10)
- W — White port (VLAN 20)
- GW — Gray + White port (Trunk)
- VLAN 10 (Gray) frame
- VLAN 20 (White) frame

Key Point:
VLANs are isolated. Packets are forwarded only within the same VLAN across bridges/switches.

VLAN Tagging

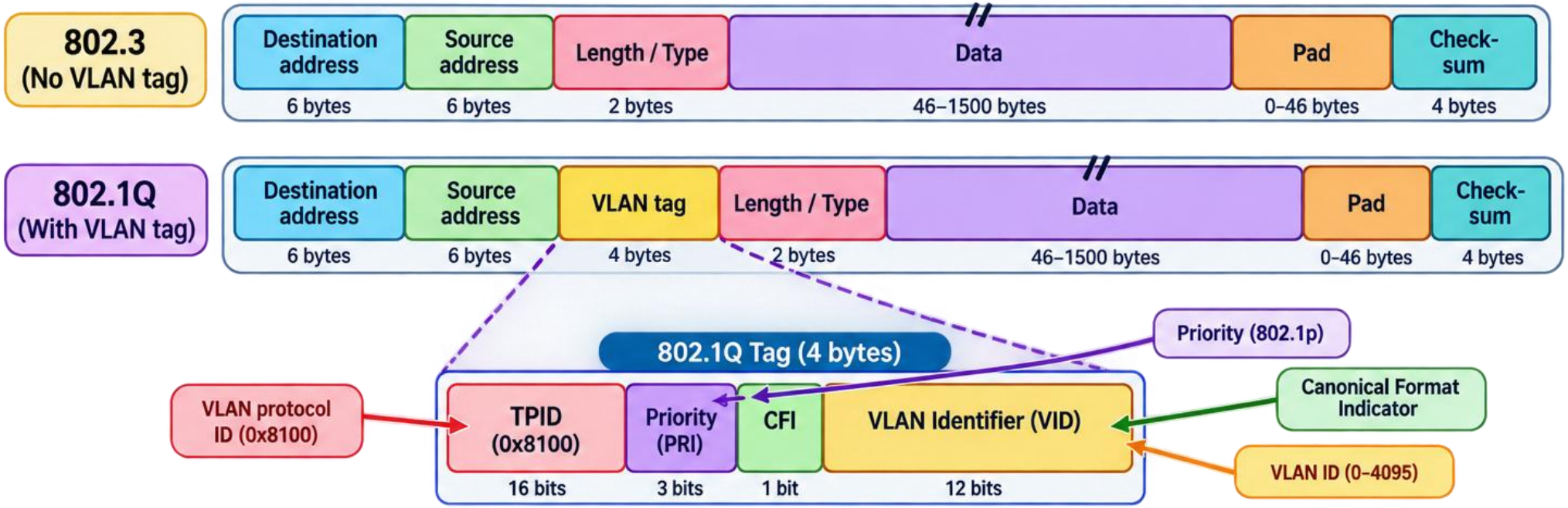
- It is placed between the source MAC and the EtherType fields

<table>
<tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>18 ...</td></tr>
<tr><td colspan="6" rowspan="2">Destination address</td><td colspan="6" rowspan="2">Source address</td><td colspan="4">VLAN tag</td><td rowspan="2">EtherType</td><td rowspan="2">Payload</td></tr>
<tr><td colspan="2">0x8100</td><td colspan="2">TCI</td></tr>
</table>

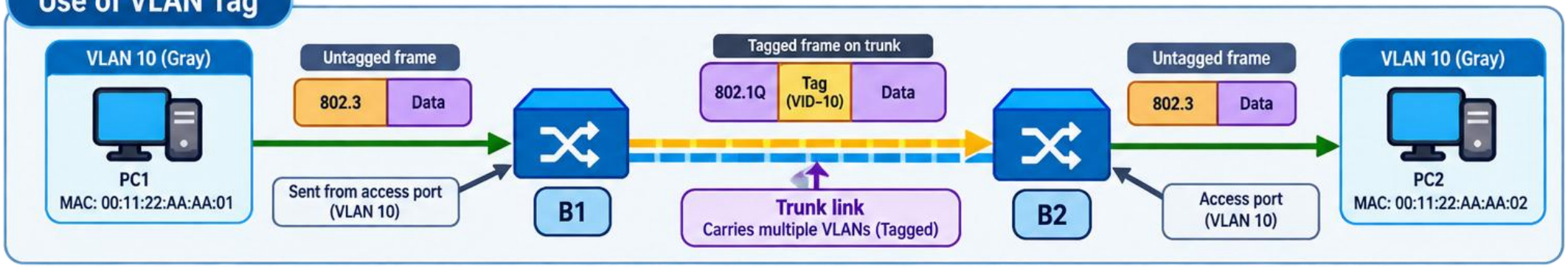
- The first two bytes of the tag contain the TPID (tag protocol identifier), which is defined to be equal to 0x8100
- The remaining two bytes contain the TCI (tag control information), of which 12 bits correspond to the VID and 4 bits contain metadata used for quality of service management.
- Each 802.1Q VLAN is identified by a 12-bit integer called a VID (VLAN Identifier) in the range 1 to 4094 inclusive.
- The values 0 and 4095 are reserved and should not be used.
- The first VLAN, with a VID of 1, is the default VLAN to which ports are presumed to belong

IEEE 802.1Q VLAN Tagging in Ethernet

Adds a 4-byte tag to identify the VLAN



Use of VLAN Tag





Key Point:
VLAN tag lets switches identify which VLAN a frame belongs to, so multiple VLANs can share the same link.

V.U.

Classic Ethernet (IEEE 802.3)

Solving $\sum_{j=0}^{\infty}(j * A\,(1-A)^{(j-1)})$

- $Mean = \sum_{j=0}^{\infty}(j * A\,(1-A)^{(j-1)})$
- Since the term evaluates to 0 when j=0, we can start the summation at j=1 without changing the result
- To solve this infinite series, we can use the derivative of a standard geometric series. Let x = (1-A). We can factor out A from the summation:

  $Mean = A\sum_{j=1}^{\infty} j \cdot x^{j-1}$
- The sum of an infinite geometric series for |x| < 1 is: $\sum_{j=0}^{\infty} x^j = \frac{1}{1-x}$
- If we take the derivative of both sides with respect to x: $\frac{d}{dx}\sum_{j=0}^{\infty} x^j = \frac{d}{dx}\left(\frac{1}{1-x}\right)$

  $\sum_{j=1}^{\infty} j \cdot x^{j-1} = \frac{1}{(1-x)^2}$
- Now, substitute x = (1-A) back into the differentiated equation:

  $\sum_{j=1}^{\infty} j \cdot (1-A)^{j-1} = \frac{1}{(1-(1-A))^2} = \frac{1}{A^2}$
- Finally, multiply by the A we factored out at the beginning: $Mean = A \cdot \left(\frac{1}{A^2}\right) = \frac{1}{A}$

End of Chapter 4

The MAC Sublayer